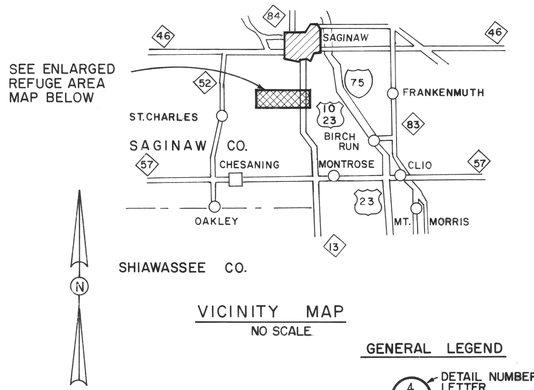
SHIAWASSEE MAINTENANCE FACILITY

SHIAWASSEE NATIONAL WILDLIFE REFUGE

SAGINAW COUNTY, MICHIGAN

U.S. FISH AND WILDLIFE SERVICE

TWIN CITIES, MINNESOTA



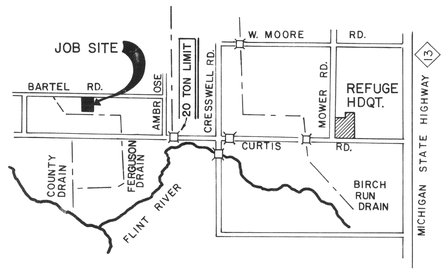
VICINITY MAP
NO SCALE

ABBREVIATIONS

@	AT
AGG.	AGGREGATE
APPROX.	APPROXIMATE
BLDG.	BUILDING
BLK.	BLOCK
B.M.	BENCH MARK
BTU/HR.	BRITISH THERMAL UNIT PER HOUR
C.F.M.	CUBIC FEET PER MINUTE
℄	CENTER LINE
COL.	COLUMN
CONC.	CONCRETE
CONN.	CONNECTION
CONT.	CONTINUOUS
CYL.	CYLINDER
DIA. or ϕ	DIAMETER
DR.	DRIVE
EA.	EACH
ELEC.	ELECTRICAL
ELEV.	ELEVATION
EXIST.	EXISTING
EXT.	EXTERIOR
F.E.	FIRE EXTINGUISHER
FIN.	FINISH
FL.	FLOOR
FT.	FEET
GA.	GAUGE
GAL.	GALLON
G.P.M.	GALLONS PER MINUTE
GRD.	GROUND
HDWE.	HARDWARE
HORIZ.	HORIZONTAL
H.P.	HORSE POWER
LB.	POUND
MAINT.	MAINTENANCE
MANUF.	MANUFACTURER
MBH.	THOUSAND BRITISH THERMAL UNITS PER HOUR
MECH.	MECHANICAL
MICH.	MICHIGAN
MIN.	MINIMUM
N.I.C.	NOT IN CONTRACT
No.	NUMBER
O.C.	ON CENTER
O.D.	OUTSIDE DIAMETER
OH.	OVERHEAD
℄ or PL.	PLATE
P.P.	POWER POLE
PROJ.	PROJECTION
PROP.	PROPOSED
PSI	POUNDS PER SQUARE INCH
R.	RADIUS
RD.	ROAD
REINF	REINFORCED
SAN.	SANITARY
SEC.	SECTION
SHT.	SHEET
T & B	TOP & BOTTON
TEX. I-II	TEXTURE I-II
TYP.	TYPICAL
U.H.	UNIT HEATER
VERT.	VERTICAL
W/	WITH
WWF	WELDED WIRE FABRIC
YACC	YOUNG ADULT CONSERVATION CORP

INDEX OF DRAWINGS

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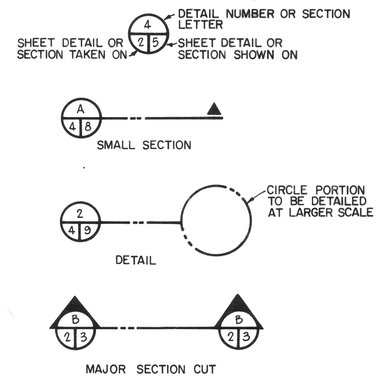


REFUGE AREA MAP
NO SCALE

LEGEND OF SYMBOLS

- GRANULAR FILL
- CONCRETE (IN SECTION)
- STEEL
- WELDED WIRE FABRIC
- EARTH
- BLOCK OR BRICK (IN SECTION)
- GROUT
- FINISH WOOD
- PLYWOOD
- INSULATION

GENERAL LEGEND



JOHNSON & ANDERSON INC.
CONSULTING ENGINEERS

REVISIONS	DATE	DRAWN

UNITED STATES DEPARTMENT OF THE INTERIOR
FISH AND WILDLIFE SERVICE
REGION 3 — TWIN CITIES, MINN.
BRANCH OF ENGINEERING

SUBMITTED — REGIONAL ENGINEER
REVIEWED
RECOMMENDED — REGIONAL DIRECTOR
SAFETY APPROVAL DATE: Gerald C. Mohl 3-10-81

SHIAWASSEE NATIONAL WILDLIFE REFUGE

SHIAWASSEE MAINTENANCE BUILDING
TITLE, INDEX & VICINITY MAP

DESIGNED W. DUNN | DRAWN W. DUNN DATE 5-22-80 | CHECKED W.D.

DRAWING NO. 3R-MICH.-458-74

DIKE ROAD TO AMBROSE RD.

+95.4 +95.3

℄ EXIST. OVERHEAD ELEC. LINES

P.P. W/GUY WIRES
B.M. TT-16
NAIL N. SIDE OF POLE
ELEV. 592.7

WOODED AREA

B
2 3
N.I.C.

265'±

42' 93'-4" 51'-8" 24' 54'

150'

P. POLE W/ GUY WIRE

ELEC. METER POLE W/ UNDERGROUND CONN.

98.5

FENCE N.I.C.

24' GATE N.I.C.

EXISTING STORAGE
Fl. Elev.
100.50
BUILDING
(N. I. C.)

5 SPACES @ 10'

4" CONC. WALK

B-1 B-2 B-3 B-4

62'-8" 67' 64' 56'-4" 260'±

10' 10' 3 1/2"

EXISTING 8" RIED CABLE

SEPTIC TILE FIELD N.I.C.

LIFT STATION

96.9

4" WD. CONC. PAD

ELEC. FEEDER

13'-4"

EXIST. 1" WATER LINE

4" CONC. WALK

99.0

TOP OF SLOPE

3 on 1 SLOPE

EXIST. Y.A.C.C. BLDG. Fl. ELEV. 100.00

Ex. 500 Gal. L.P. Tank

L.P. GASLINE

3 SPACES @ 10'

Septic Tank

MAINTENANCE
Fl. Elev.
100.50
BUILDING
RE-SREAD EXISTING GRAVEL SEE NOTE 3A

550 GAL. 4'Ø WASTE OIL STORAGE TANK

CONC. ENTRANCE APRON SEE DETAIL 8 SHEET 5 (TYP.)

EXISTING 6" CRUSHED AGG. BASE FOR ENTR. DR. & COURTYARD

12 R

9 SPACES @ 10'

T/B

B/B

CONCRETE PARKING RAIL SEE DETAIL SHT. 3 N.I.C.

3:1 Slope

8'± HIGH MOUND

10' 32' 80' 72' 56'

250'±

E. PUMP ISLAND N.I.C.

4' GATE

3-1000 GAL. FUEL TANKS

19'±

40'

30' 5' 20' 15'

70' 250'±

52' 45'

30' R

WOODED AREA

LOW AREA

OPEN

FLAT

EDGE OF WOODED AREA

0 10 20 30 60
SCALE: 1"=30'

SEPTIC TILE FIELD N.I.C.

LIFT STATION
4" INLET INV. 95.6
1½" OUTLET INV. 99.2

2'x2' DISTRIBUTION BOX (SEE DETAIL SHT. 3
1½" INV. E. 99.0
4" INV. S. 98.5
4" INV. N. 98.4

N.I.C.
Septic Tank

100'-4" SAN @ 1.00 %
(1)

EXISTING Y.A.C.C. BLDG.

PROPOSED MAINTENANCE BUILDING

INV. 96.6

1000 GAL. SEPTIC TANK (SEE DETAIL SHT. 3)
INV. 97.5

NOTES:

1.) WASTE OIL STORAGE TANK AND SEPTIC TANK TO BE DESIGNED & CONSTRUCTED FOR MDOT H 20 LOADING.

2.) ELEVATIONS SHOWN ARE BASED ON ASSUMED ELEVATION OF 100.00 AT EXISTING FINISHED FLOOR OF Y.A.C.C. BLDG.

3.) 104.5 INDICATES PROPOSED ELEVATION

4.) REMOVE EXISTING 6" GRAVEL WITHIN MAINTENANCE BLDG. CONSTRUCTION LIMITS AND RE-SPREAD PER SPEC SECTION 02512.

5.) INDICATES DIRECTION OF PROPOSED SURFACE DRAINAGE

6.) BUILDING EL. 100.00 = 596.64 MSL

7.) SOIL BORING LOGS & SOILS REPORT ARE INCLUDED IN SPECIFICATIONS.

SHIAWASSEE NATIONAL WILDLIFE REFUGE

JOHNSON & ANDERSON, INC.
CONSULTING ENGINEERS
PONTIAC, MICHIGAN

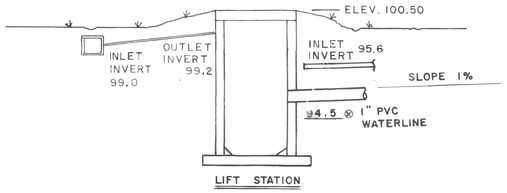
REVISION	DATE	DESCRIPTION	BY
(1)	8-30-83	"AS BUILT"	GAD

SITE PLAN

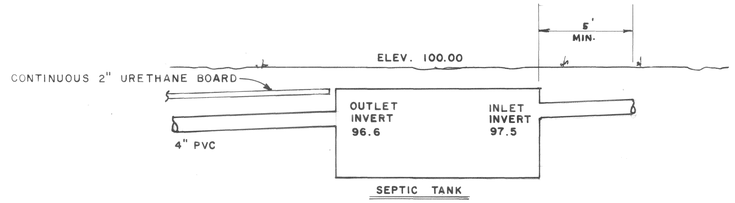
SECONDARY HEADQUARTERS SITE MAINTENANCE COMPLEX
SHIAWASSEE NATIONAL WILDLIFE REFUGE, SAGINAW CO., MICH.

DESIGNED JJA | DRAWN RAI | DATE 5-22-80 | CHECKED W.D

DRAWING NO. 3R—MICH.— 458—75

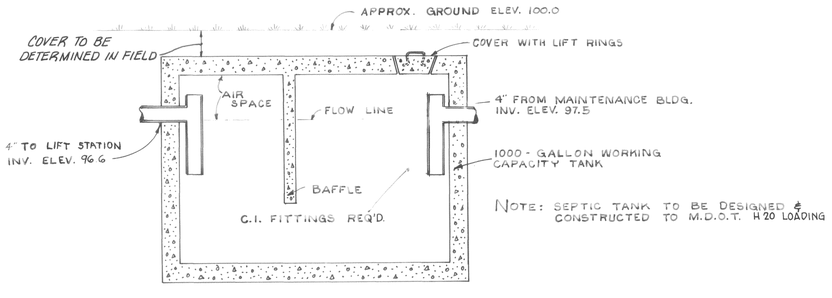


LIFT STATION



SEPTIC TANK

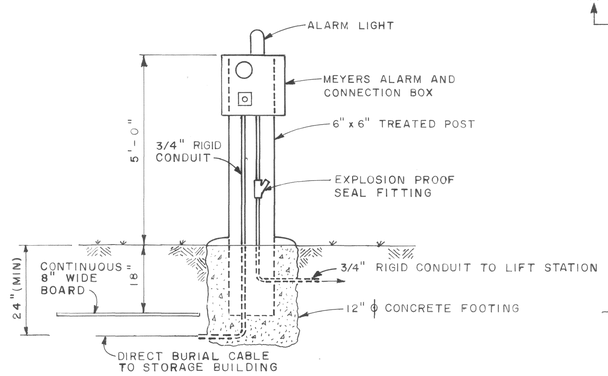
NO SCALE



NOTE: SEPTIC TANK TO BE DESIGNED & CONSTRUCTED TO M.D.O.T. H 20 LOADING

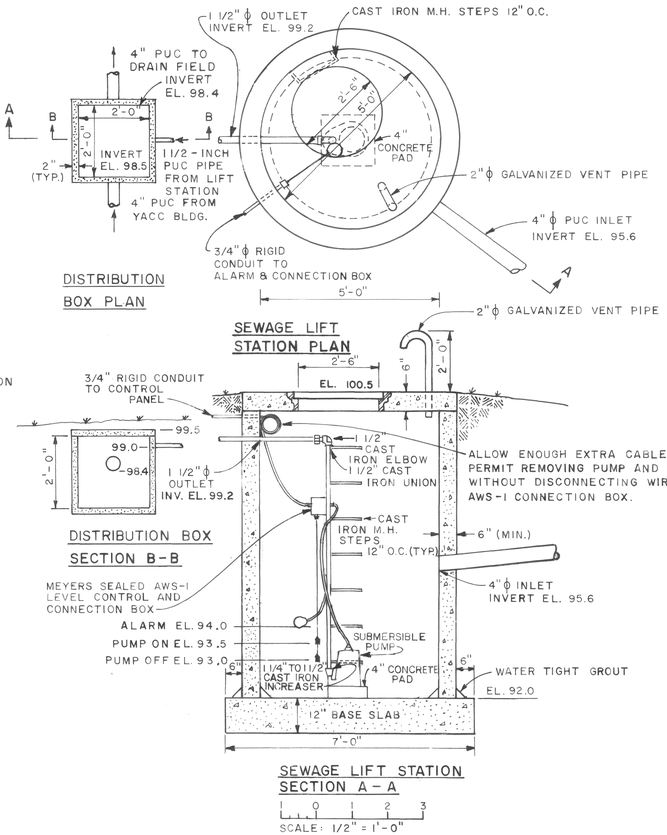
SEPTIC TANK DETAIL

NOT TO SCALE



ALARM & CONNECTION BOX

NOT TO SCALE



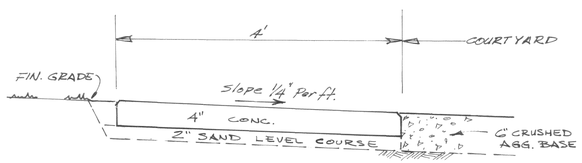
DISTRIBUTION BOX PLAN

DISTRIBUTION BOX SECTION B-B

SEWAGE LIFT STATION PLAN

SEWAGE LIFT STATION SECTION A-A

SCALE: 1/2" = 1'-0"



TYPICAL 4' CONC. WALK

DETAIL

SCALE: 1" = 1'-0"

SHIAWASSEE NATIONAL WILDLIFE REFUGE

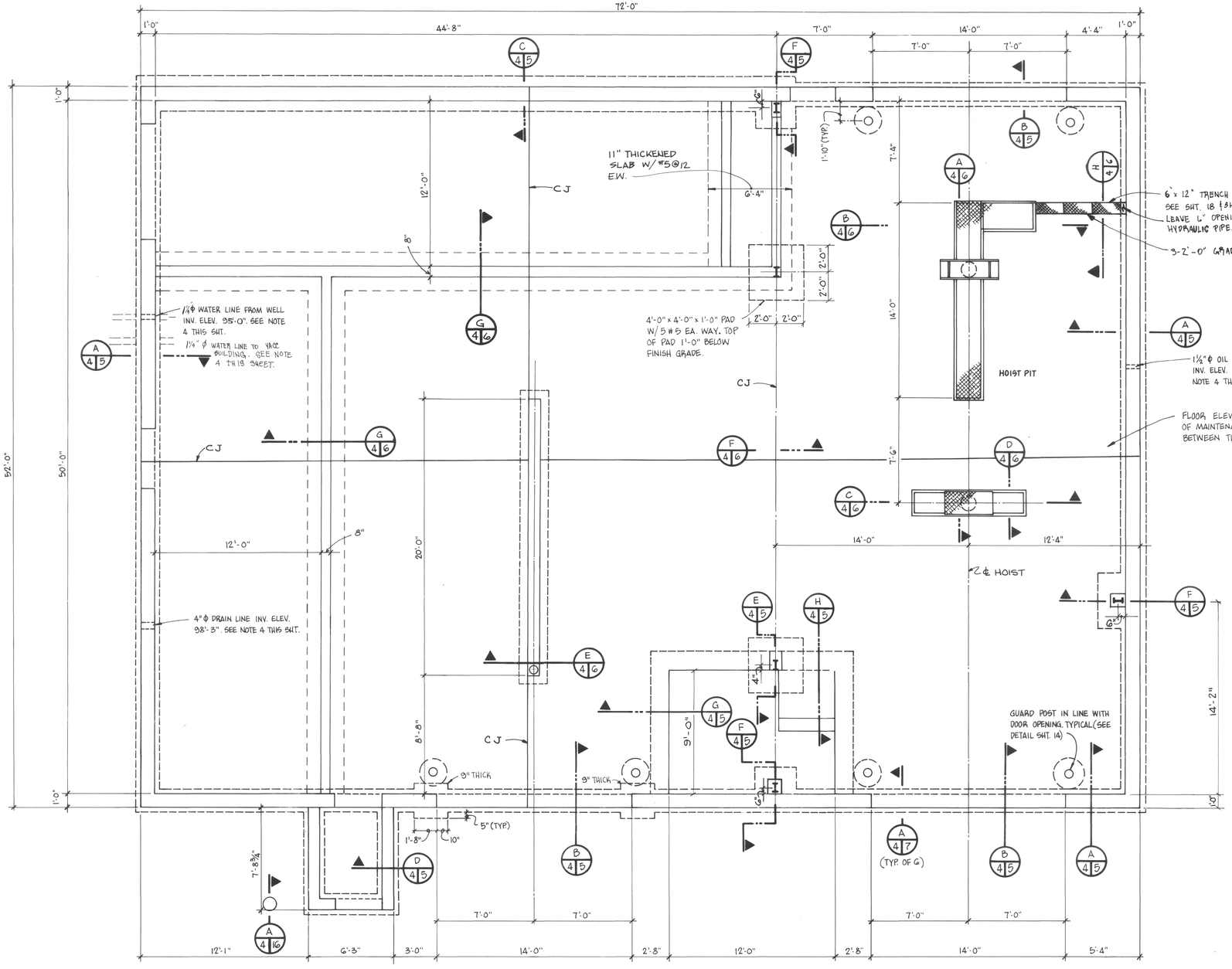
REVISION	DATE	DESCRIPTION	BY

SITE - SECTIONS & DETAILS

DESIGNED JJK | DRAWN JJK | DATE 5-22-80 | CHECKED W.D.

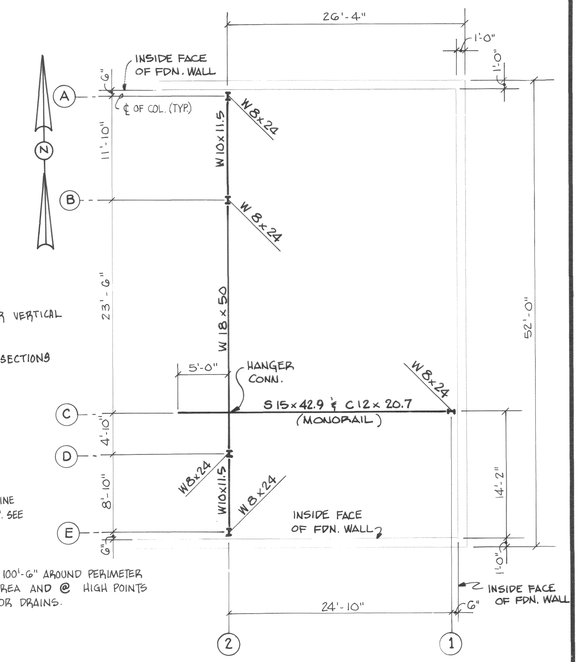
DRAWING NO. 3R-MICH-458-76

JOHNSON & ANDERSON, INC.
CONSULTING ENGINEERS
PONTIAC, MICHIGAN



MAINTENANCE BUILDING FOUNDATION PLAN

SCALE: ¼" = 1'-0"



STEEL FRAMING PLAN

SCALE: ⅛" = 1'-0"

NOTE: SEE DRAWINGS 7, 8, 15 & 16 FOR STRUCT. STEEL SECTIONS & DETAILS RELATED TO ABOVE PLAN.

NOTES:

1. C.J. DENOTES CONTROL JOINT. USE SECTION F 4|6
2. CONSTRUCTION JOINT SECTION J 4|6 CAN BE USED AS AN ALTERNATE TO CONTROL JOINT SECTION F 4|6.
3. FLOOR HARDENER & SEALER SHALL BE APPLIED TO CONCRETE FLOOR PER SPECIFICATIONS.
4. SEE SHT 18 FOR PIPING LAYOUT. COORDINATE LOCATION WITH MECHANICAL CONTRACTOR. VERIFY INVERTS.

0 1' 2' 4' 6' 8'

SCALE: ¼" = 1'-0"

REVISION	DATE	DESCRIPTION	BY

MAINTENANCE BUILDING FOUNDATION PLAN (AND STEEL FRAMING PLAN)

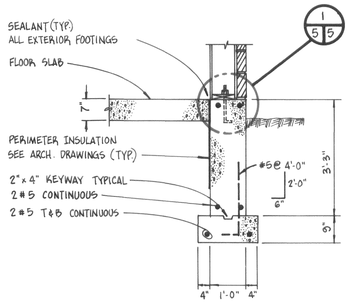
DESIGNED JMC | DRAWN W. DUNN DATE 5-22-80 | CHECKED DAC

DRAWING NO. 3R—MICH.—458—77

JOHNSON & ANDERSON, INC.
CONSULTING ENGINEERS
PONTIAC, MICHIGAN

SHIAWASSEE NATIONAL WILDLIFE REFUGE

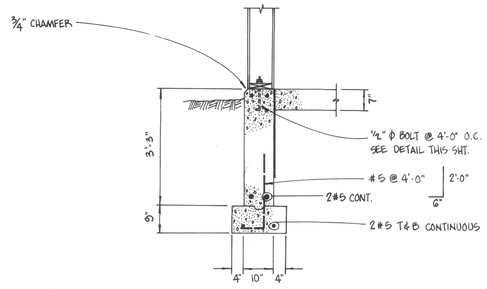
SHEET 4 OF 23



TYPICAL WALL FOOTING

SCALE: 1/2" = 1'-0"

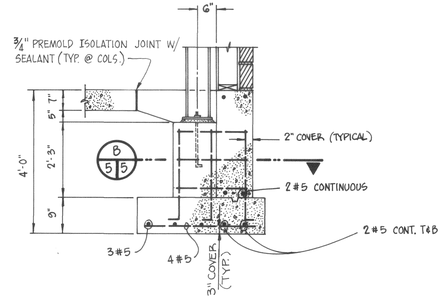
A 4|5



SECTION @ VESTIBULE

SCALE: 1/2" = 1'-0"

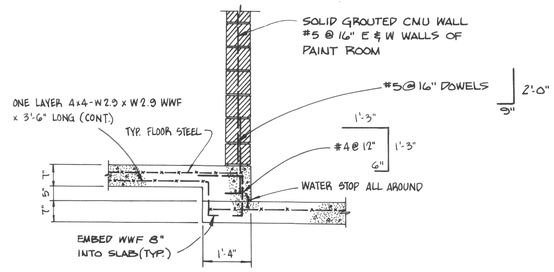
D 4|5



SECTION @ COLUMN FOOTING

SCALE: 1/4" = 1'-0"

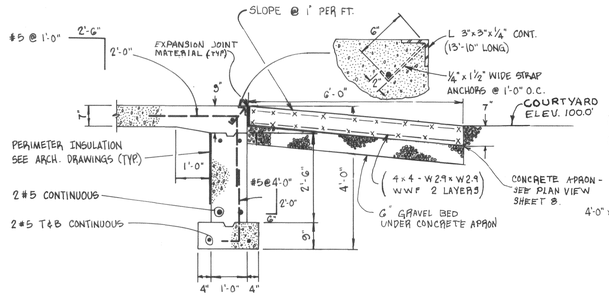
F 4|5



SECTION @ PAINT ROOM

SCALE: 1/2" = 1'-0"

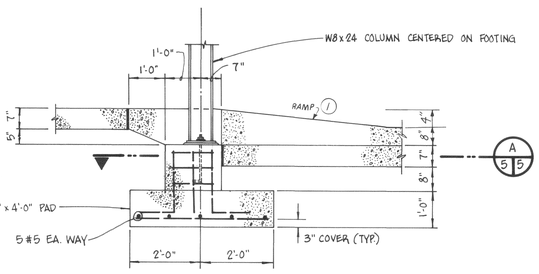
G 4|5



WALL FOOTING @ O.H. DOORS

SCALE: 1/2" = 1'-0"

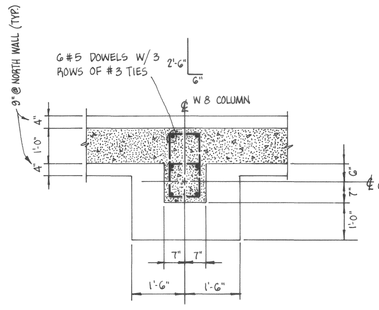
B 4|5



SECTION @ COLUMN FOOTING

SCALE: 1/2" = 1'-0"

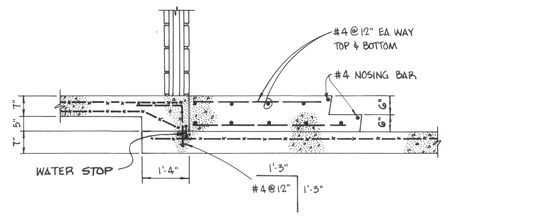
E 4|5



PLAN @ COLUMN FOOTING

SCALE: 1/2" = 1'-0"

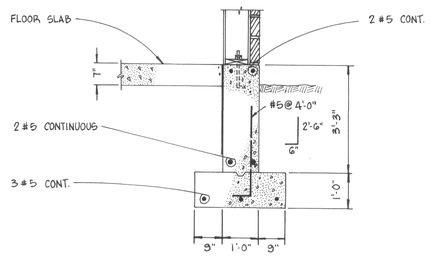
B 5|5



SECTION @ PAINT ROOM

SCALE: 1/2" = 1'-0"

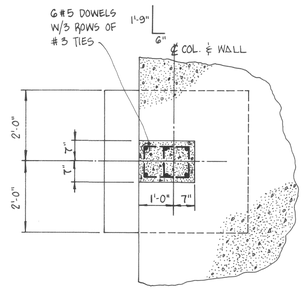
H 4|5



WALL FOOTING @ MEZZANINE STORAGE

SCALE: 1/2" = 1'-0"

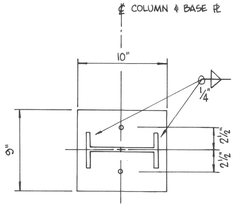
C 4|5



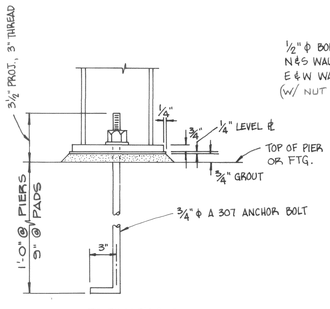
PLAN @ COLUMN FOOTING

SCALE: 1/2" = 1'-0"

A 5|5



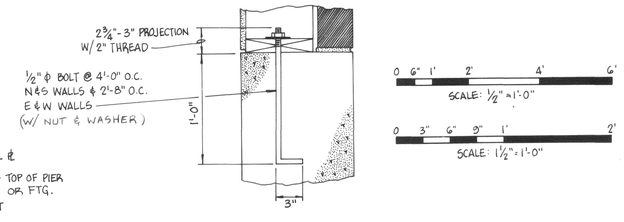
PLAN



ELEVATION

COLUMN BASE DETAIL - TYPICAL

NO SCALE



SILL PLATE ANCHOR DETAIL

SCALE: 1 1/2" = 1'-0"

I 5|5

REVISION	DATE	DESCRIPTION	BY
(1)	8-30-83	"AS BUILT"	GAO

MAINTENANCE BUILDING SECTIONS & DETAILS

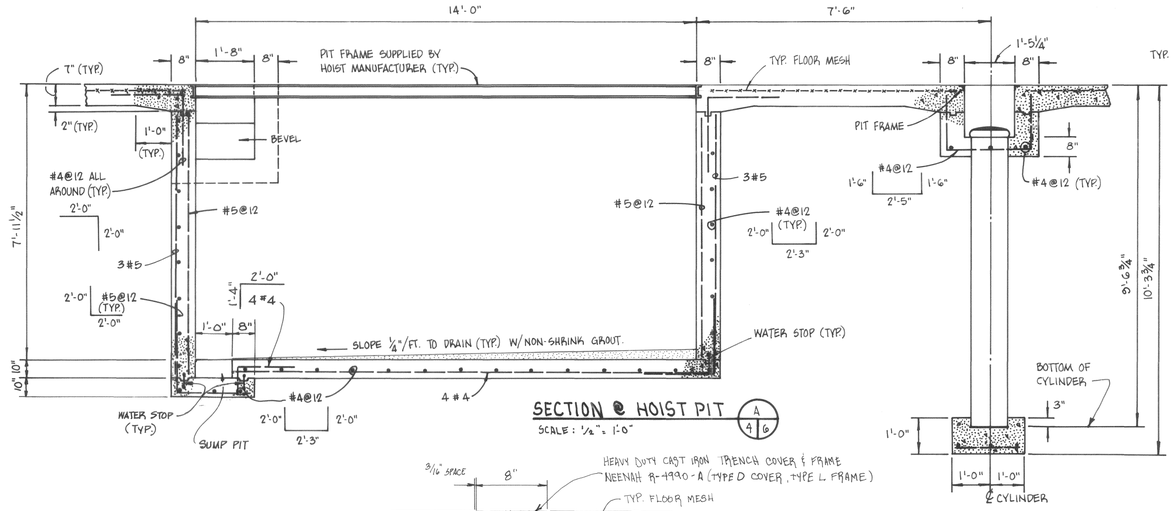
DESIGNED JMC | DRAWN W DUNN DATE 5-22-80 | CHECKED DHC

DRAWING NO. 3R—MICH.—458—78

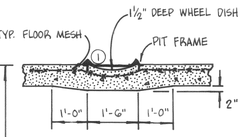
JOHNSON & ANDERSON, INC.
CONSULTING ENGINEERS
PONTIAC, MICHIGAN

SHIAWASSEE NATIONAL WILDLIFE REFUGE

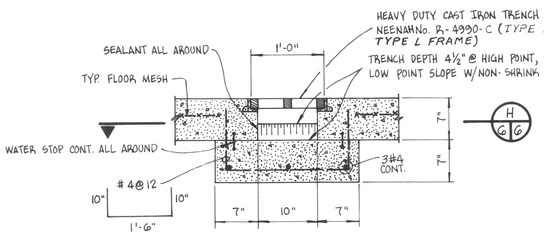
SHEET 5 OF 23



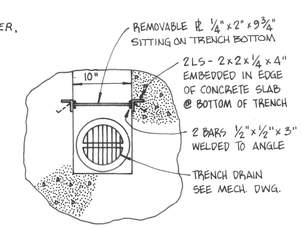
SECTION @ HOIST PIT
SCALE: ½" = 1'-0"
A 4 6



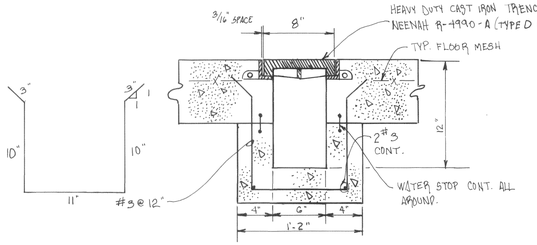
SECTION
SCALE: ½" = 1'-0"
D 4 6



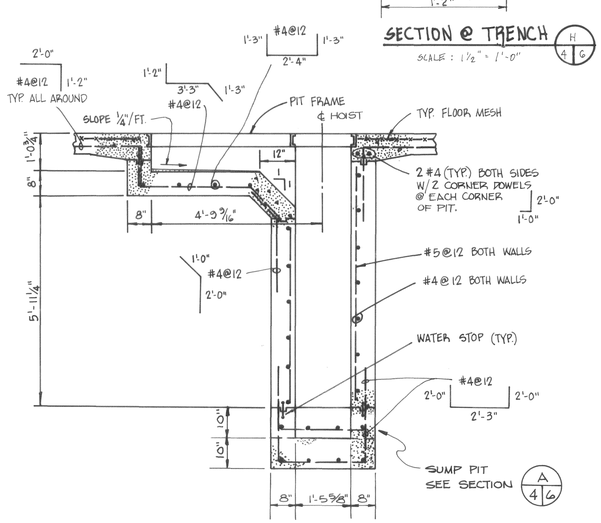
SECTION @ TRENCH
SCALE: 1" = 1'-0"
E 4 6



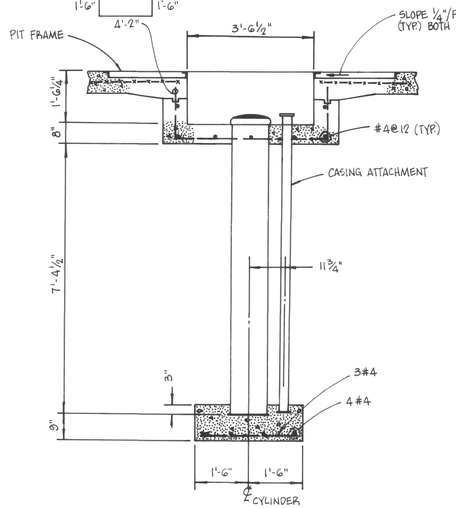
SECTION @ TRENCH
H 6 6



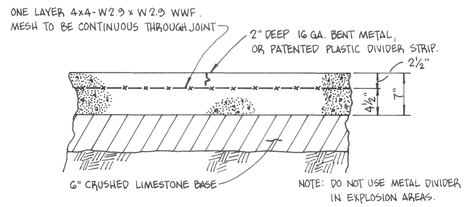
SECTION @ TRENCH
SCALE: 1½" = 1'-0"
H 4 6



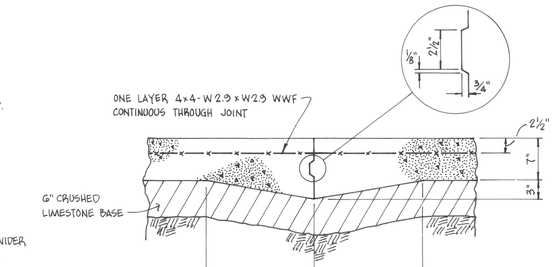
SECTION
SCALE: ½" = 1'-0"
B 4 6



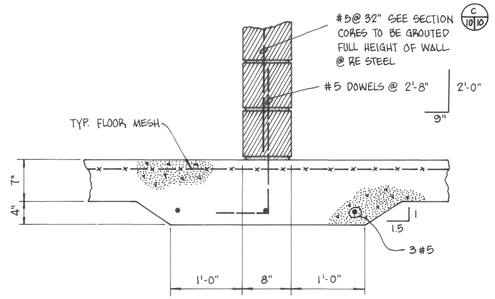
SECTION
SCALE: ½" = 1'-0"
C 4 6



TYPICAL FLOOR SLAB & CONTROL JOINT
SCALE: 1" = 1'-0"
F 4 6



TYPICAL CONSTRUCTION JOINT
SCALE: 1" = 1'-0"
J 4 6



SECTION @ THICKENED SLAB
SCALE: 1" = 1'-0"
G 4 6

NOTE:
1. ALL HOIST PIT DIMENSIONS TO BE VERIFIED WITH HOIST MANUFACTURER.
2. HOIST PIT EXCAVATION, CONCRETING PROCEDURE AND INSTALLATION SHALL CONFORM TO INSTRUCTIONS OF THE HOIST MANUFACTURER.

0 6" 1' 2' 4' 6'
SCALE: ½" = 1'-0"

0 6" 1' 2' 3'
SCALE: 1" = 1'-0"

JOHNSON & ANDERSON, INC.
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PONTIAC, MICHIGAN

REVISION	DATE	DESCRIPTION	BY
1	9-9-83	"AS BUILT"	GAO

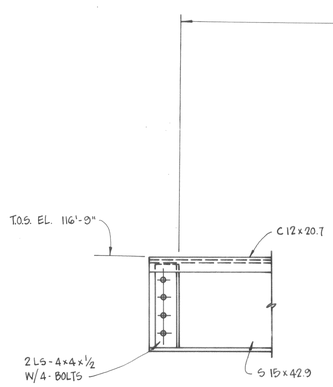
MAINTENANCE BUILDING MISC. CONCRETE SECTIONS

DESIGNED JMC | DRAWN W. DUNN DATE 5-22-80 | CHECKED DKC

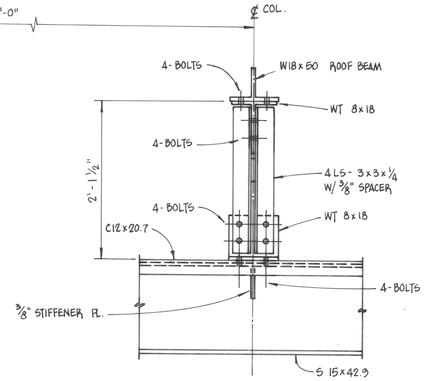
DRAWING NO. 3R—MICH.—458—79

SHIAWASSEE NATIONAL WILDLIFE REFUGE

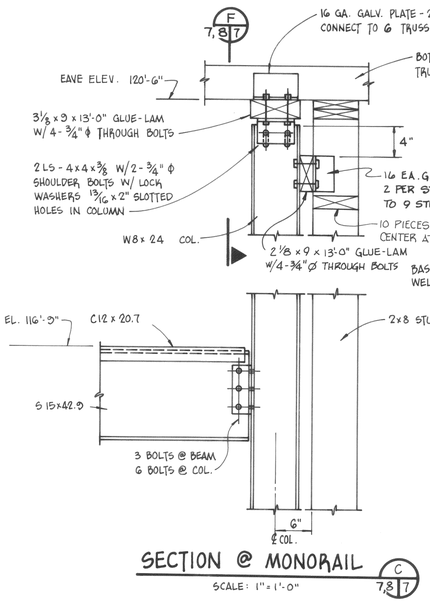
SHEET 6 OF 23



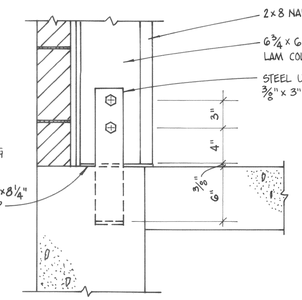
SECTION @ MONORAIL (A 8|7)
SCALE: 1" = 1'-0"



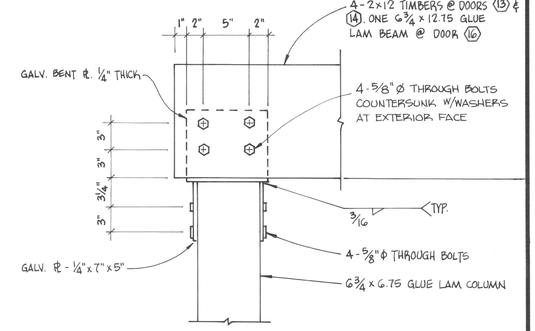
SECTION @ MONORAIL (B 8|7)
SCALE: 1" = 1'-0"



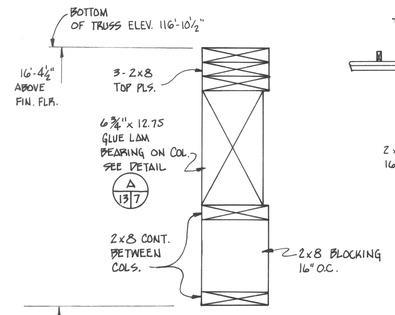
SECTION @ MONORAIL (C 7,8|7)
SCALE: 1" = 1'-0"



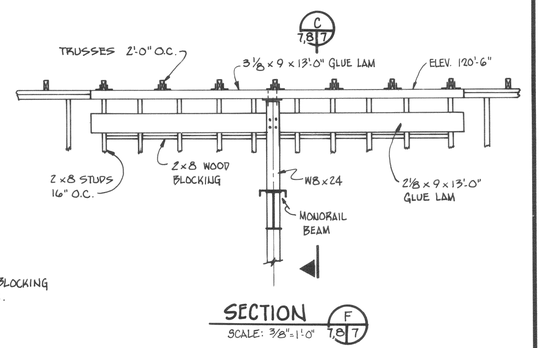
SECTION (A 4|7)
SCALE: 1½" = 1'-0"



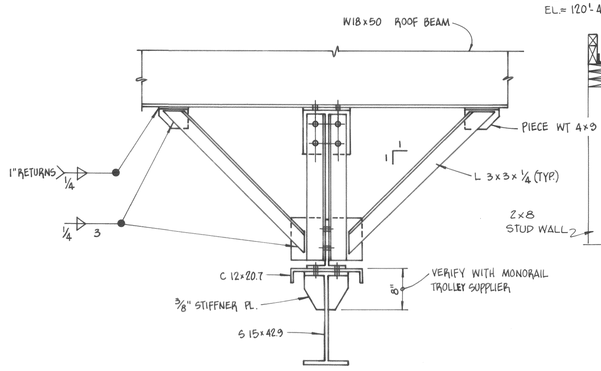
DETAIL (A 13|7)
SCALE: 1½" = 1'-0"



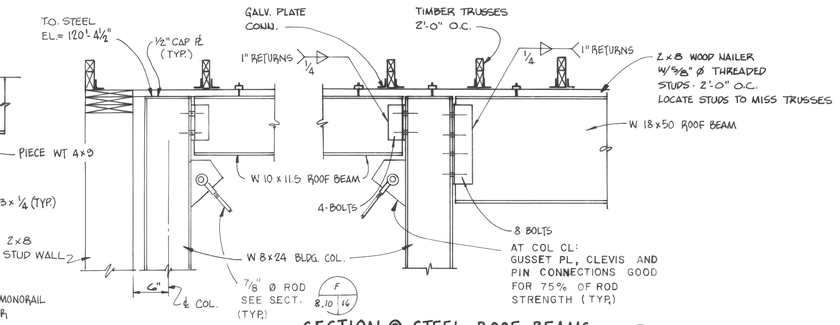
DETAIL @ DOOR (16) (B 13|7)
SCALE: 1½" = 1'-0"



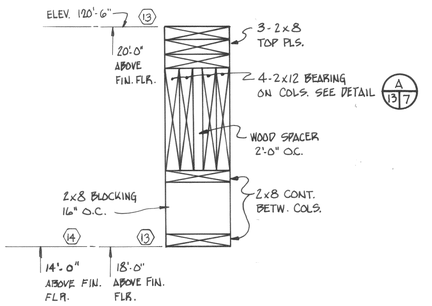
SECTION (F 7,8|7)
SCALE: ⅜" = 1'-0"



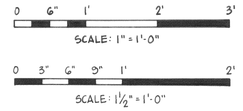
SECTION @ MONORAIL (D 8|7)
SCALE: 1" = 1'-0"



SECTION @ STEEL ROOF BEAMS (E 8|7)
SCALE: 1" = 1'-0"



DETAIL @ DOORS (13), (14) (C 13|7)
SCALE: 1½" = 1'-0"



NOTES:

ALL BOLTS FOR STRUCTURAL STEEL CONNECTIONS SHALL BE ¾" ϕ A 325 HIGH STRENGTH BOLTS UNLESS OTHERWISE NOTED.

ELEVATIONS BASED ON FINISHED FLOOR ELEVATION = 100'-0"

ALL TIMBER FITTINGS AND BOLTS THIS SHEET SHALL BE GALVANIZED.

REVISION	DATE	DESCRIPTION	BY

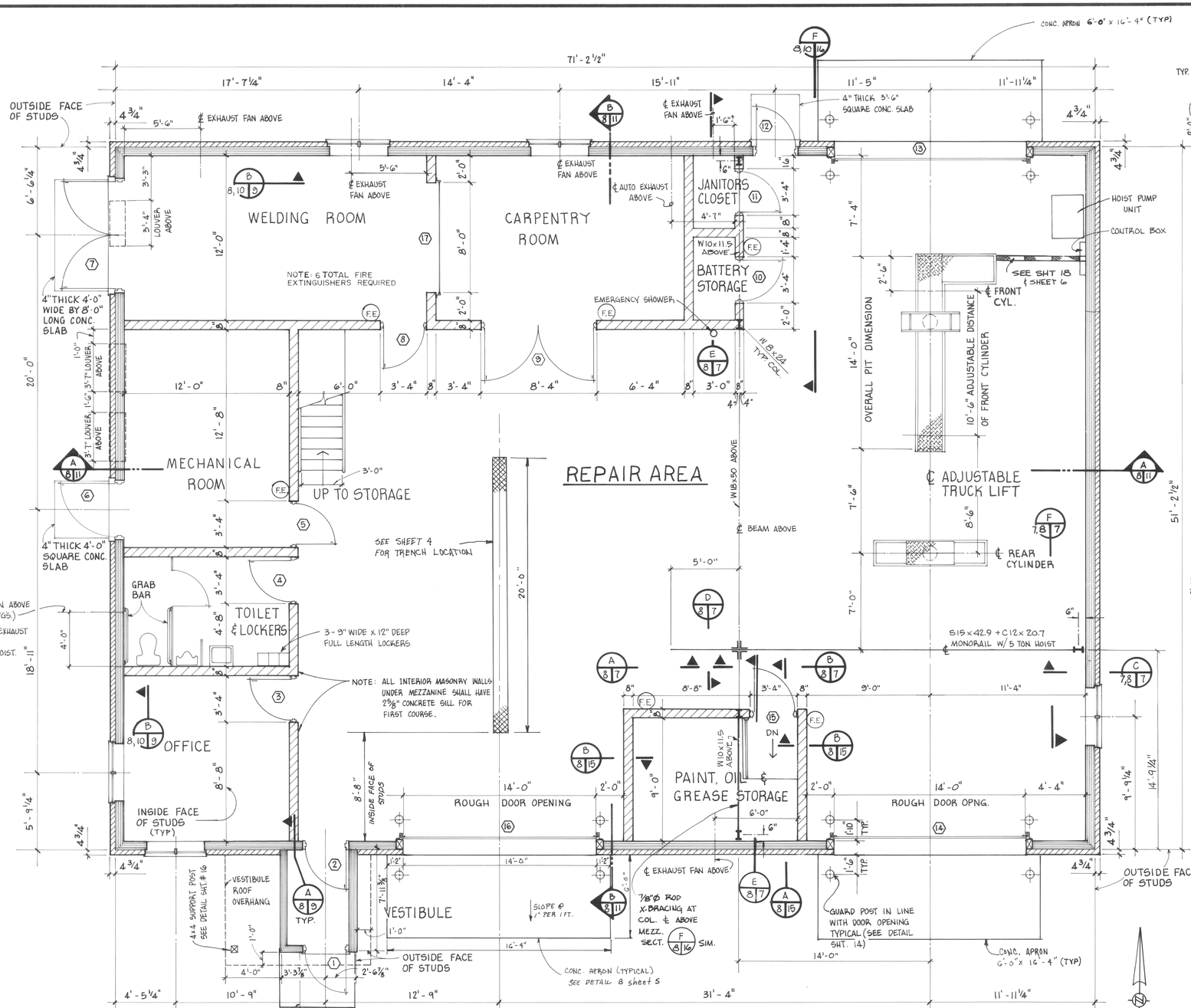
STRUCTURAL FRAMING SECTIONS

	DESIGNED JMC	DRAWN W DUNN DATE 5-22-80	CHECKED
		DRAWING NO 3R — MICH. — 458 — 80	

JOHNSON & ANDERSON, INC.
CONSULTING ENGINEERS
PONTIAC, MICHIGAN

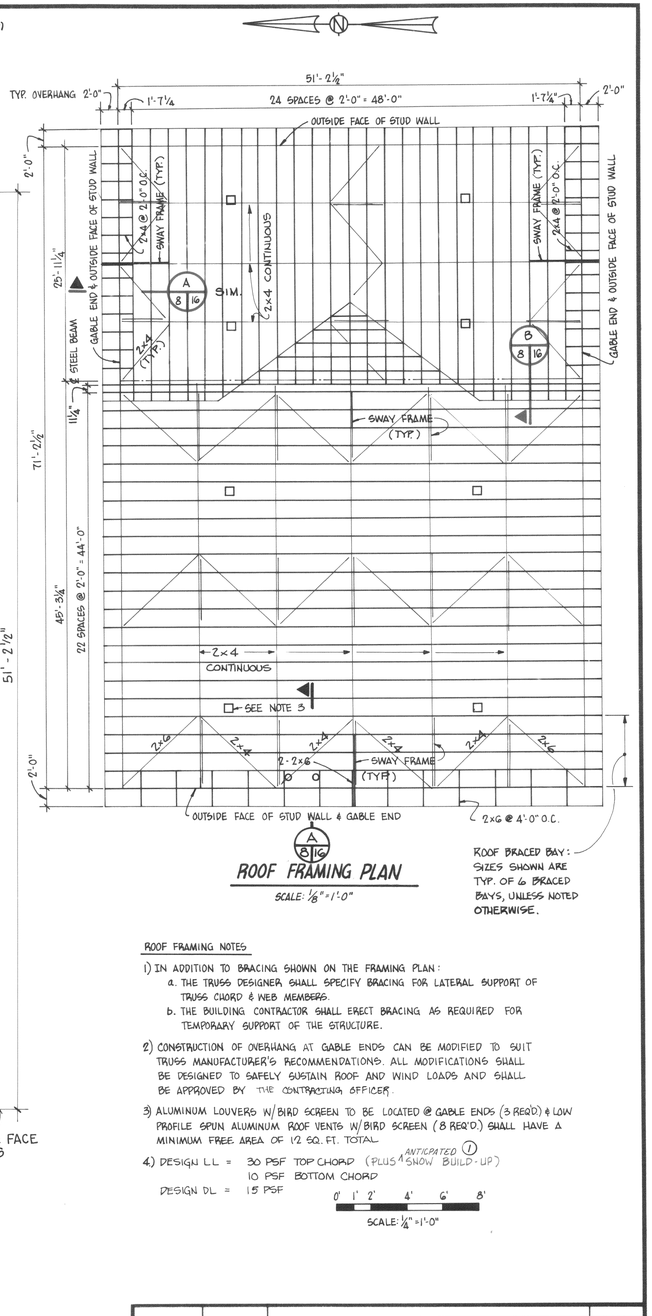
SHIAWASSEE NATIONAL WILDLIFE REFUGE

SHEET 7 OF 23



PLAN- MAINTENANCE BUILDING

SCALE 1/4" = 1'-0"



ROOF FRAMING PLAN

SCALE: 1/8" = 1'-0"

ROOF BRACED BAY: SIZES SHOWN ARE TYP. OF 4 BRACED BAYS, UNLESS NOTED OTHERWISE.

ROOF FRAMING NOTES

1) IN ADDITION TO BRACING SHOWN ON THE FRAMING PLAN:
   a. THE TRUSS DESIGNER SHALL SPECIFY BRACING FOR LATERAL SUPPORT OF TRUSS CHORD & WEB MEMBERS.
   b. THE BUILDING CONTRACTOR SHALL ERECT BRACING AS REQUIRED FOR TEMPORARY SUPPORT OF THE STRUCTURE.
2) CONSTRUCTION OF OVERHANG AT GABLE ENDS CAN BE MODIFIED TO SUIT TRUSS MANUFACTURER'S RECOMMENDATIONS. ALL MODIFICATIONS SHALL BE DESIGNED TO SAFELY SUSTAIN ROOF AND WIND LOADS AND SHALL BE APPROVED BY THE CONTRACTING OFFICER.
3) ALUMINUM LOUVERS W/BIRD SCREEN TO BE LOCATED @ GABLE ENDS (3 REQ'D) & LOW PROFILE SPUN ALUMINUM ROOF VENTS W/BIRD SCREEN (8 REQ'D) SHALL HAVE A MINIMUM FREE AREA OF 12 SQ. FT. TOTAL
4) DESIGN LL = 30 PSF TOP CHORD (PLUS 1 ANTICIPATED SNOW BUILD-UP)
   10 PSF BOTTOM CHORD
   DESIGN DL = 15 PSF

SCALE: ¼" = 1'-0"

REVISION	DATE	DESCRIPTION	BY
1	8-30-83	"AS BUILT"	GAO

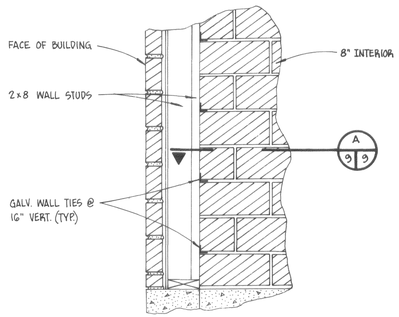
PLAN-MAINTENANCE BLDG.

DESIGNED MAG | DRAWN W. DUNN DATE 5-22-80 | CHECKED

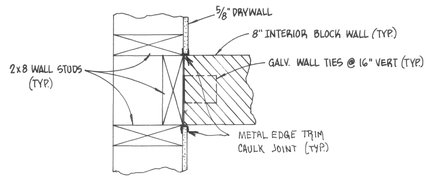
DRAWING NO. 3R-MICH.-458-81

JOHNSON & ANDERSON, INC.
CONSULTING ENGINEERS
PONTIAC, MICHIGAN

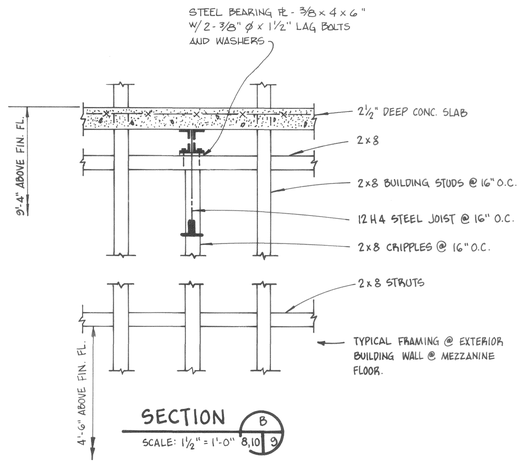
SHIAWASSEE NATIONAL WILDLIFE REFUGE



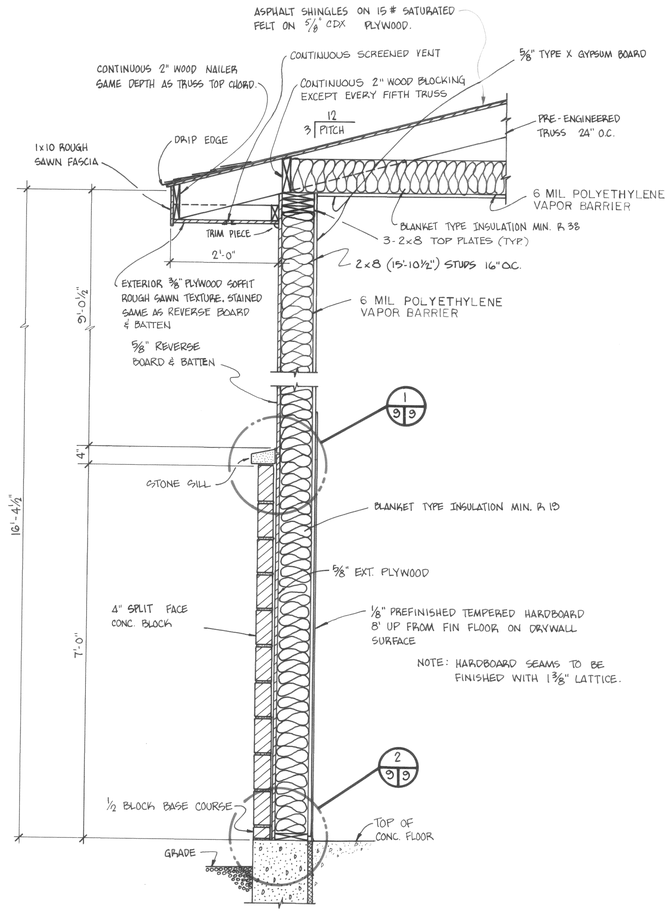
SECTION A 8 | 9



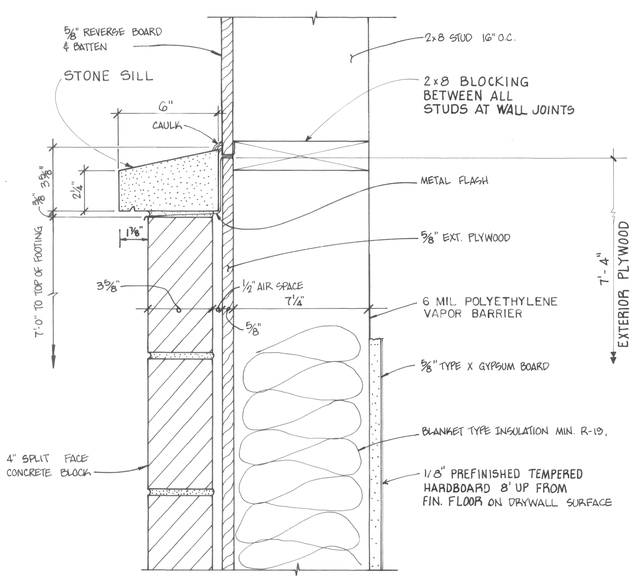
SECTION A 9 | 9



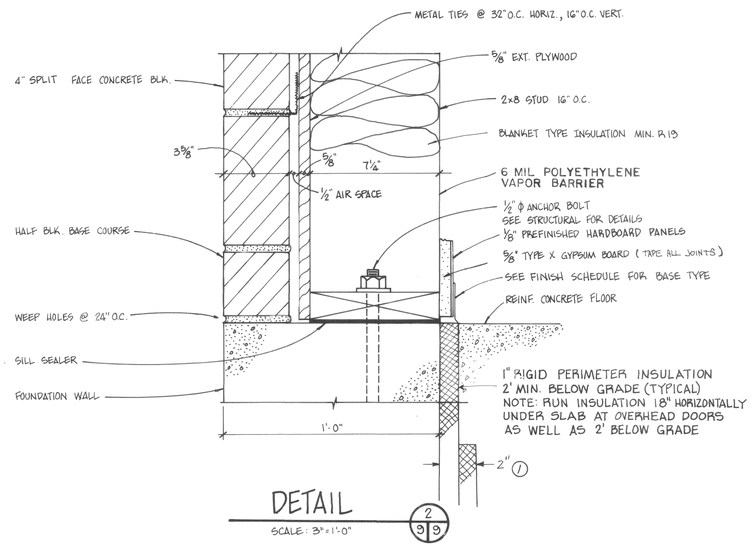
SECTION B SCALE: 1½" = 1'-0" 8,10 | 9



SECTION A SCALE: ¾" = 1'-0" 12 | 9



DETAIL SCALE: 3" = 1'-0" 1 | 9



DETAIL SCALE: 3" = 1'-0" 2 | 9

0 6" 1' 2' 3' 5'
SCALE: ¾" = 1'-0"

0 1" 2" 3" 6" 9" 1'
SCALE: 3" = 1'-0"

REVISION	DATE	DESCRIPTION	BY
(1)	8-30-83	"AS BUILT"	GAO

MAINT. BUILDING - SECTIONS & DETAILS

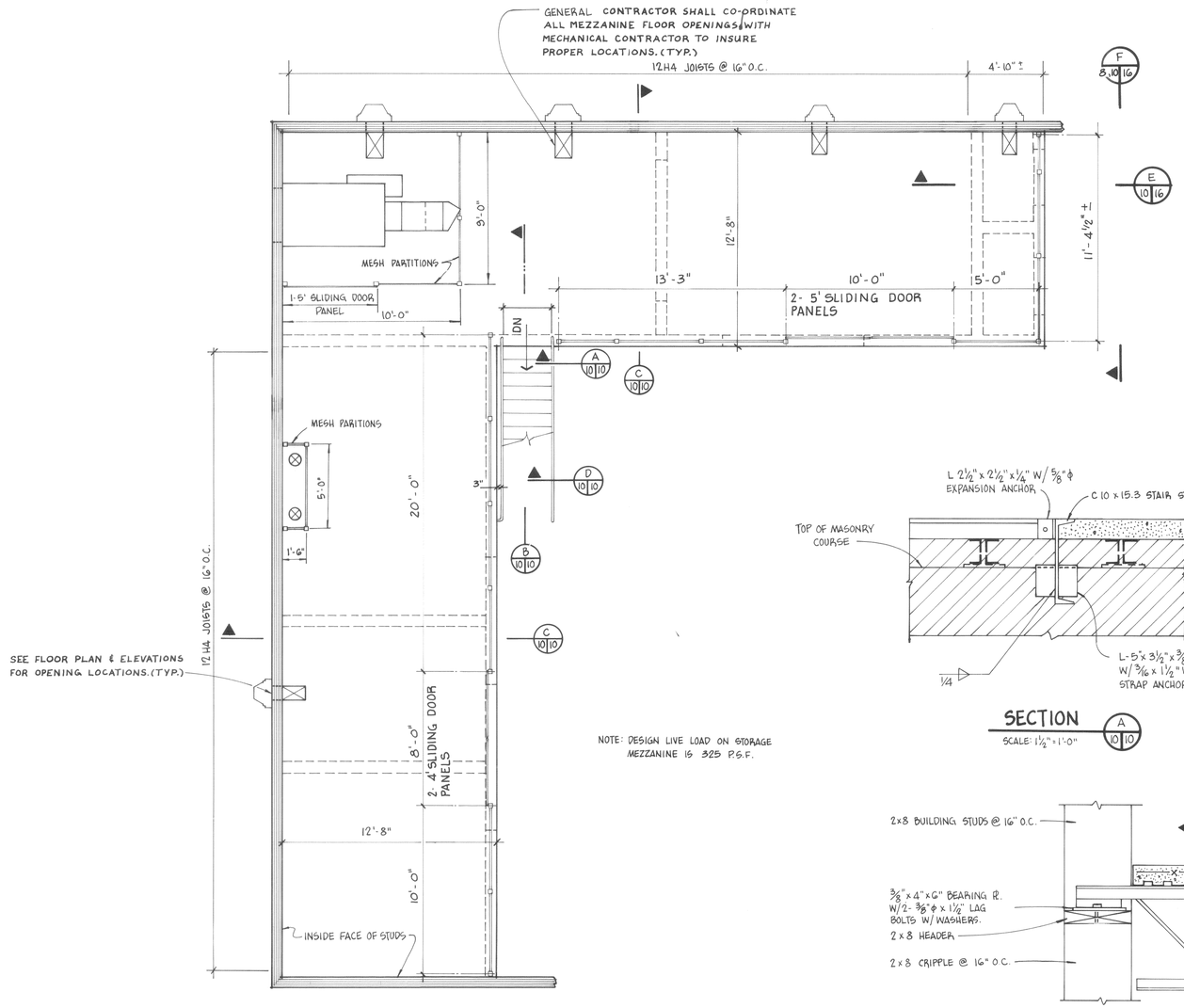
DESIGNED MAG. | DRAWN W. DUNN DATE 5-22-80 | CHECKED

DRAWING NO. 3R — MICH. — 458 — 82

JOHNSON & ANDERSON, INC.
CONSULTING ENGINEERS
PONTIAC, MICHIGAN

SHIAWASSEE NATIONAL WILDLIFE REFUGE

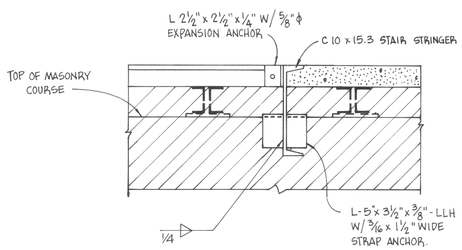
SHEET 9 OF 23



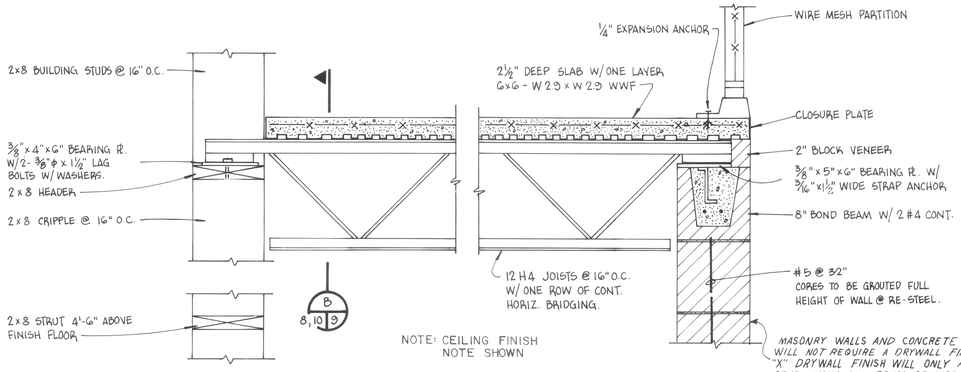
NOTE: DESIGN LIVE LOAD ON STORAGE MEZZANINE IS 325 P.S.F.

STORAGE MEZZANINE PLAN

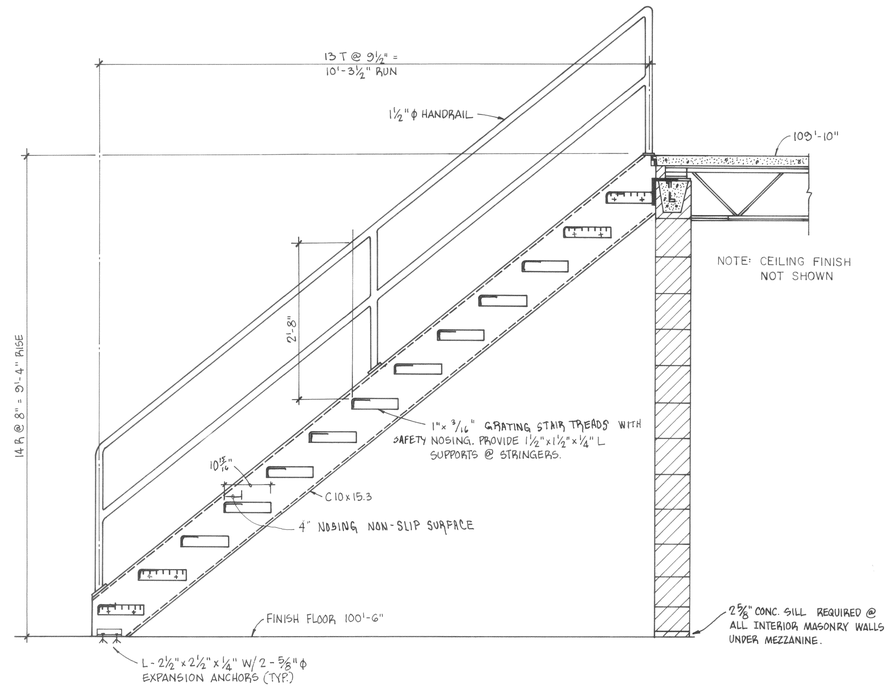
SCALE 1/4" = 1'-0"



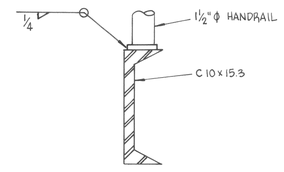
SECTION A 10|10
SCALE: 1½" = 1'-0"



NOTE: CEILING FINISH NOTE SHOWN



SECTION B 10|10
SCALE: ¾" = 1'-0"



SECTION D 10|10
SCALE: 1½" = 1'-0"

0 1' 2' 4' 6' 8'
SCALE: ¼" = 1'-0"

0 6" 1' 2' 3' 5'
SCALE: ¾" = 1'-0"

0 3" 6" 9" 1' 2'
SCALE: 1½" = 1'-0"

SECTION C 10|10
SCALE: 1½" = 1'-0"

REVISION	DATE	DESCRIPTION	BY
(1)	8-30-83	"AS BUILT"	GAO

MAINTENANCE BLDG. - STORAGE MEZZANINE PLAN SECTION & DETAILS

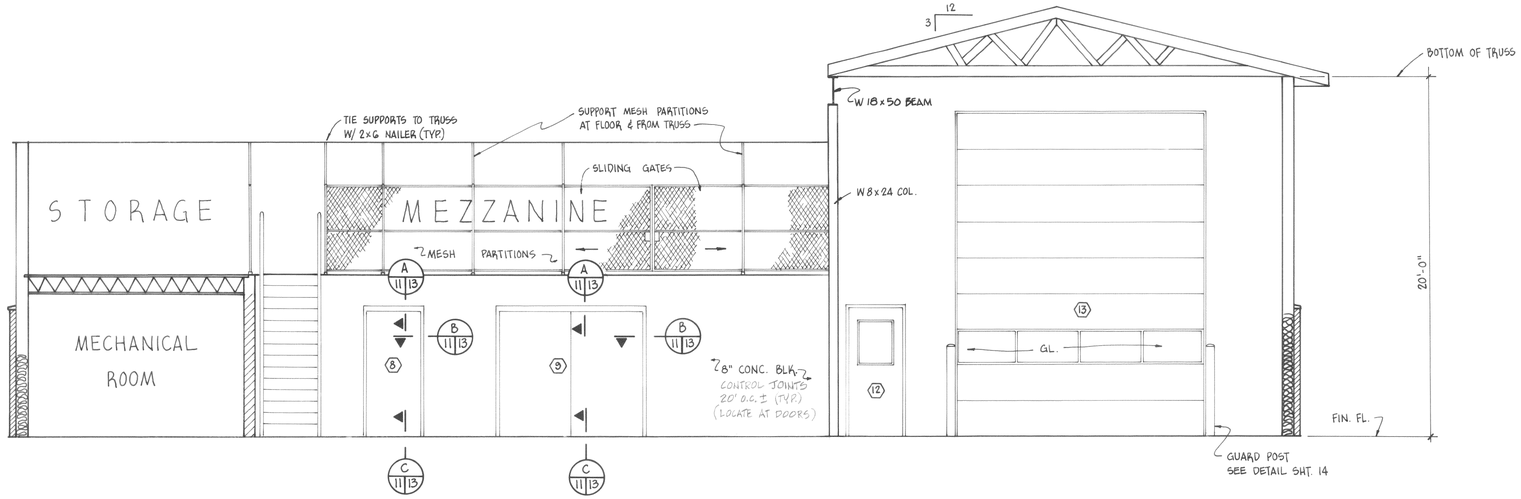
DESIGNED MAG	DRAWN MAG W. DUNN	DATE 5-22-80	CHECKED

DRAWING NO. 3R — MICH. — 458 — 83

JOHNSON & ANDERSON, INC.
CONSULTING ENGINEERS
PONTIAC, MICHIGAN

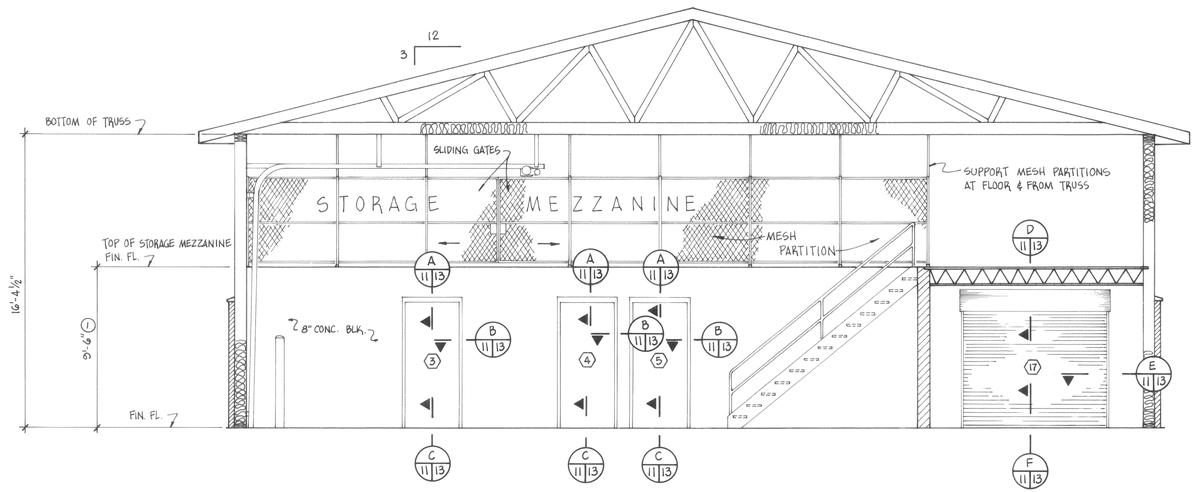
SHIAWASSEE NATIONAL WILDLIFE REFUGE

SHEET 10 OF 23



LONGITUDINAL SECTION

SCALE: 1/4" = 1'-0"



TRANSVERSE SECTION

SCALE: 1/4" = 1'-0"

0 1' 2' 4' 6' 8'

SCALE: 1/4" = 1'-0"

REVISION	DATE	DESCRIPTION	BY
1	8-30-83	"AS BUILT"	GAO

MAINTENANCE BUILDING LONGITUDINAL & TRANSVERSE SECTIONS

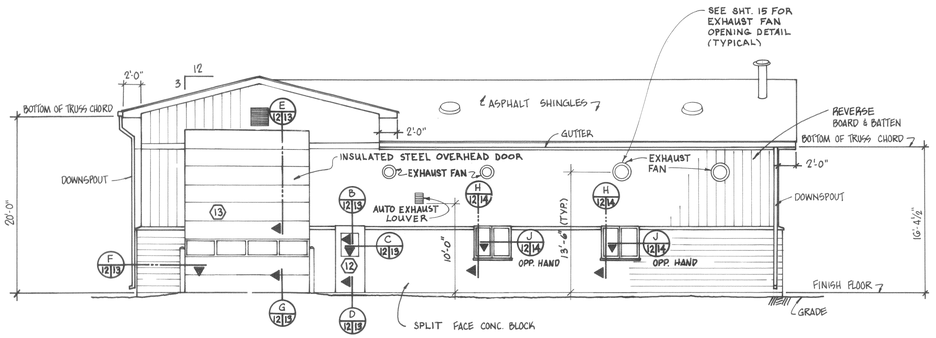
DESIGNED MAG | DRAWN W DUNN DATE 5-22-80 | CHECKED

DRAWING NO. 3R—MICH.—458—84

JOHNSON & ANDERSON, INC.
CONSULTING ENGINEERS
PONTIAC, MICHIGAN

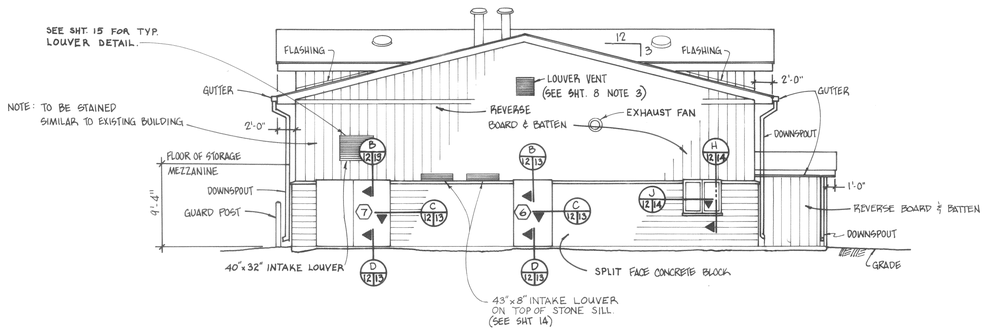
SHIAWASSEE NATIONAL WILDLIFE REFUGE

SHEET 11 OF 23



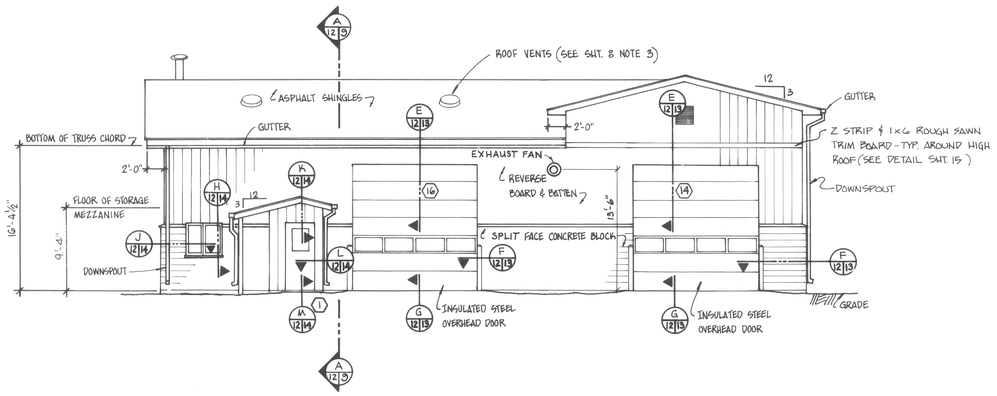
NORTH ELEVATION

SCALE: ⅛" = 1'-0"



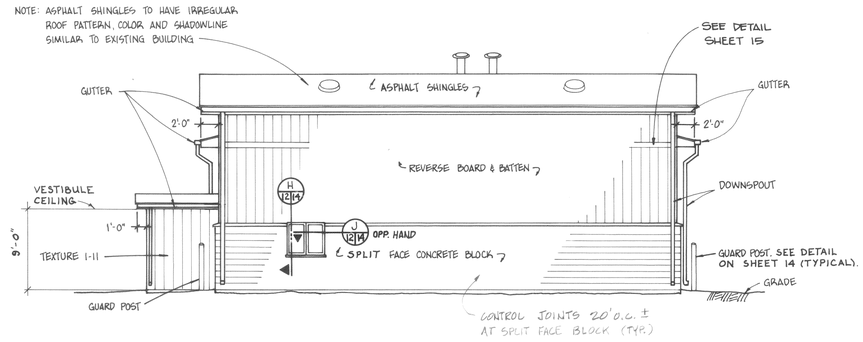
WEST ELEVATION

SCALE: ⅛" = 1'-0"



SOUTH ELEVATION

SCALE: ⅛" = 1'-0"



EAST ELEVATION

SCALE: ⅛" = 1'-0"

0' 4' 8' 12' 16'

SCALE: ⅛" = 1'-0"

REVISION	DATE	DESCRIPTION	BY

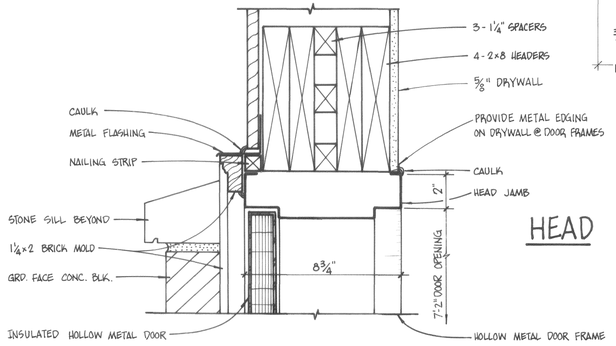
MAINTENANCE BUILDING - ELEVATIONS

DESIGNED MAG | DRAWN W. DUNN DATE 5-22-80 | CHECKED

DRAWING NO 3R — MICH. — 458 — 85

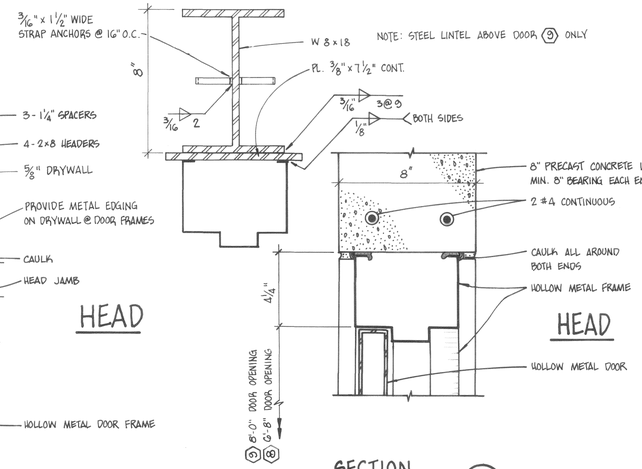
JOHNSON & ANDERSON, INC.
CONSULTING ENGINEERS
PONTIAC, MICHIGAN

SHIAWASSEE NATIONAL WILDLIFE REFUGE



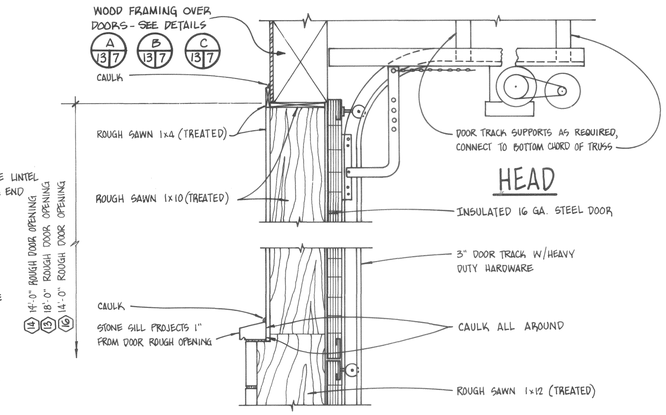
HEAD

SECTION
SCALE: 3" = 1'-0"
B 12 13



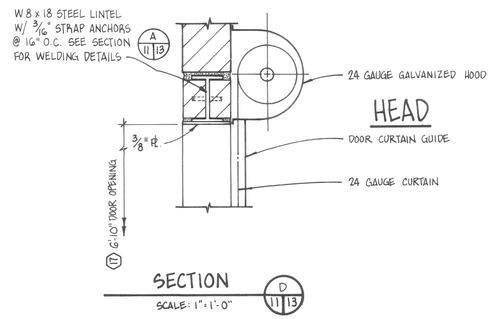
HEAD

SECTION
SCALE: 3" = 1'-0"
A 11 13

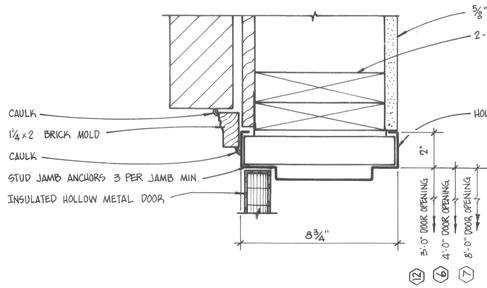


HEAD

SECTION
SCALE: 1" = 1'-0"
E 12 13

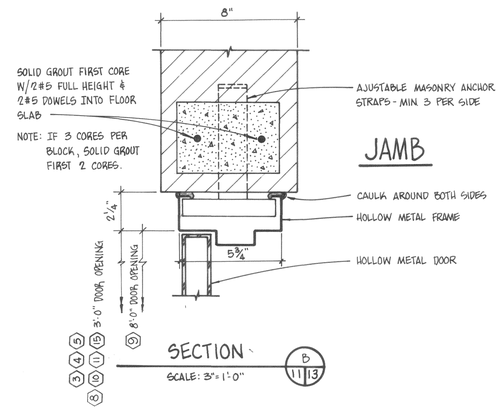


SECTION
SCALE: 1" = 1'-0"
D 11 13



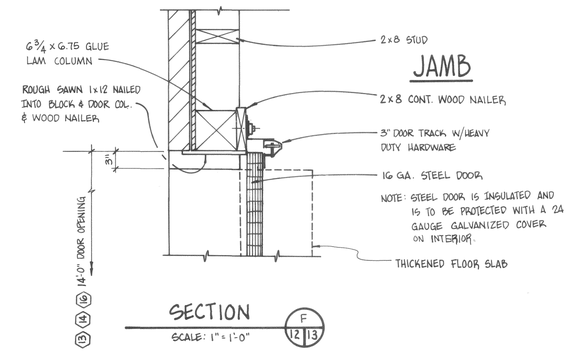
JAMB

SECTION
SCALE: 3" = 1'-0"
C 12 13



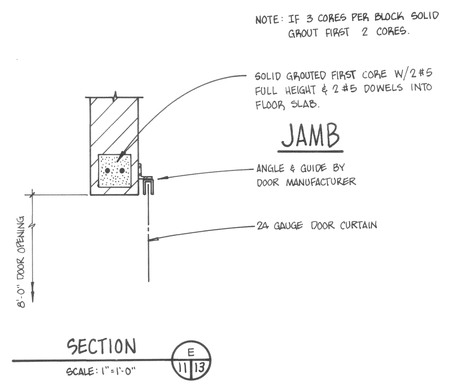
JAMB

SECTION
SCALE: 3" = 1'-0"
B 11 13

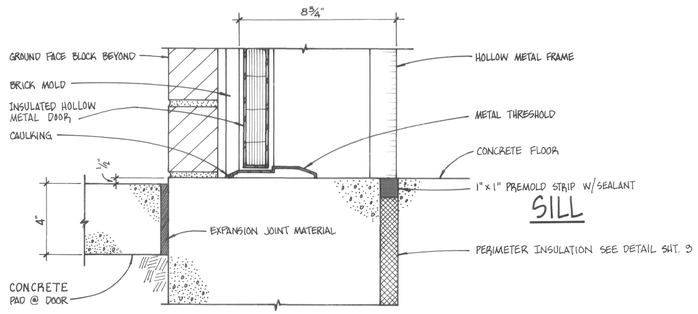


JAMB

SECTION
SCALE: 1" = 1'-0"
F 12 13

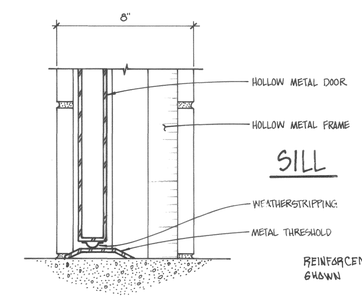


SECTION
SCALE: 1" = 1'-0"
E 11 13



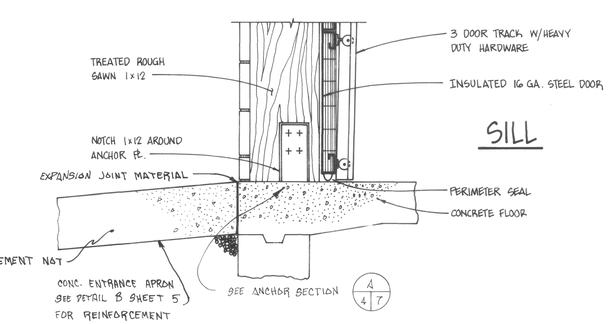
SILL

SECTION
SCALE: 3" = 1'-0"
D 12 13



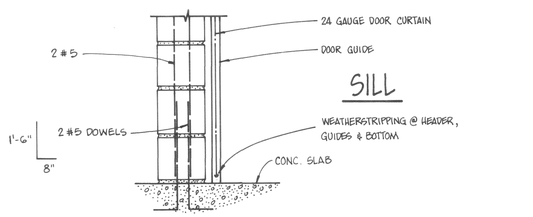
SILL

SECTION
SCALE: 3" = 1'-0"
C 11 13



SILL

SECTION
SCALE: 1" = 1'-0"
G 12 13



SECTION
SCALE: 1" = 1'-0"
F 11 13

0 6" 1' 2' 3'
SCALE: 1" = 1'-0"

0 1" 2" 3" 6" 9" 1'
SCALE: 3" = 1'-0"

REVISION	DATE	DESCRIPTION	BY

MAINTENANCE BUILDING DOOR DETAILS

DESIGNED MAG	DRAWN W. DUNN DATE 5-22-80	CHECKED
	DRAWING NO 3R — MICH. — 458 — 86	

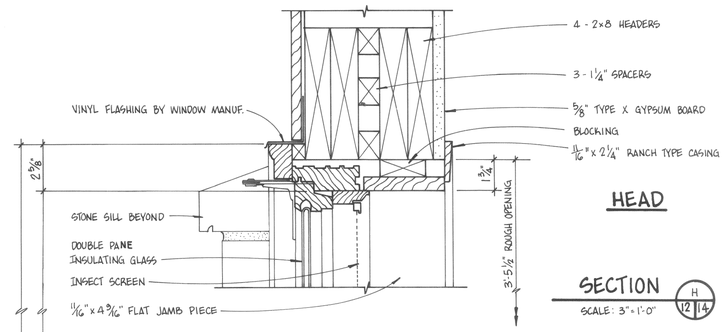
JOHNSON & ANDERSON, INC.
CONSULTING ENGINEERS
PONTIAC, MICHIGAN

SHIAWASSEE NATIONAL WILDLIFE REFUGE

SHEET 13 OF 23

GUARD POST DETAILS

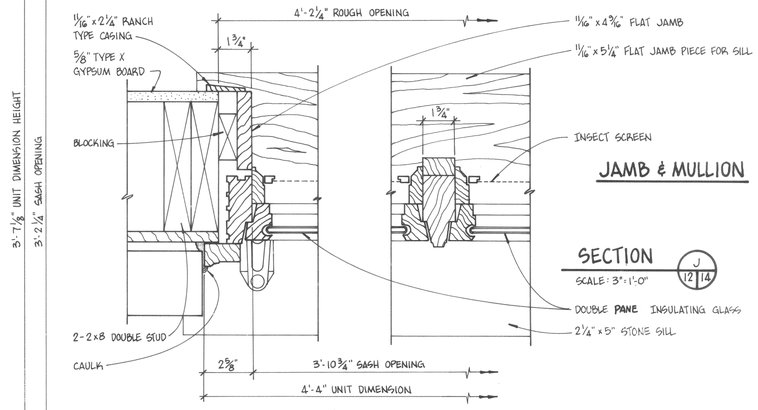
SCALE: 1" = 1'-0"



HEAD

SECTION H 12|14

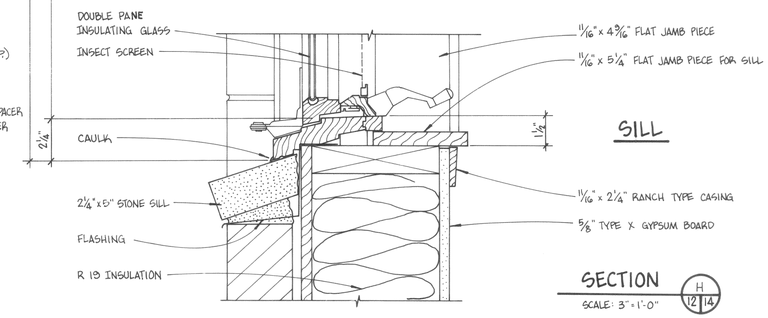
SCALE: 3" = 1'-0"



JAMB & MULLION

SECTION J 12|14

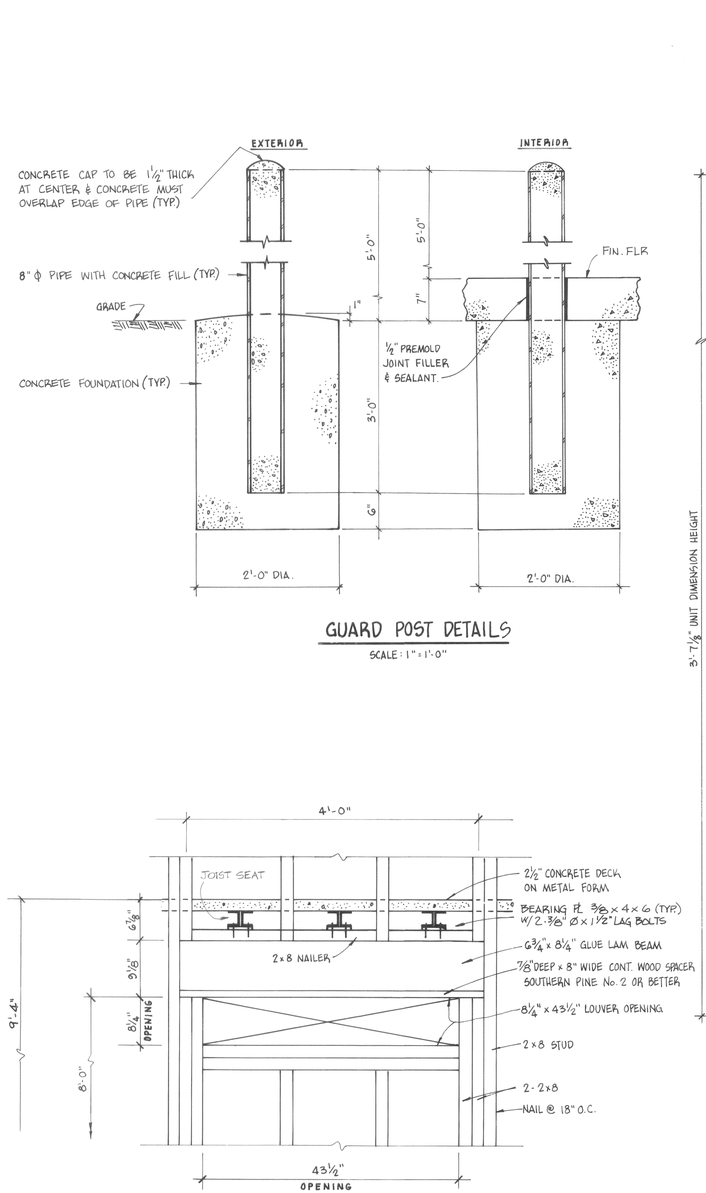
SCALE: 3" = 1'-0"



SILL

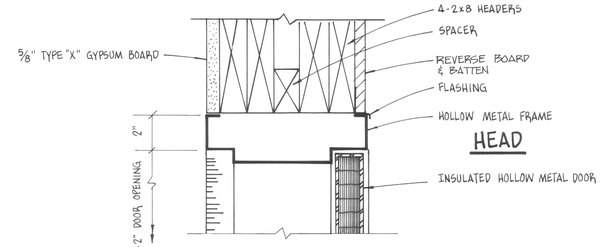
SECTION H 12|14

SCALE: 3" = 1'-0"



INTAKE LOUVER FRAMING DETAIL

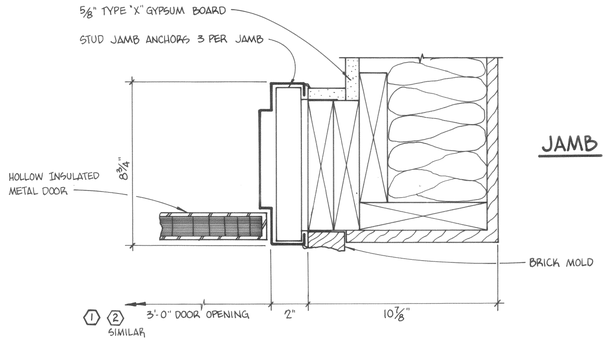
NO SCALE



HEAD

SECTION K 12|14

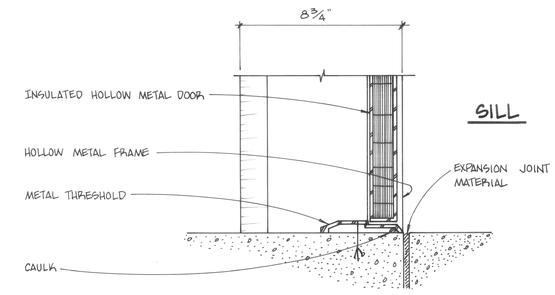
SCALE: 3" = 1'-0"



JAMB

SECTION L 12|14

SCALE: 3" = 1'-0"



SILL

SECTION M 12|14

SCALE: 3" = 1'-0"

CASEMENT WINDOW DETAILS

0 6" 1' 2' 3'

SCALE: 1" = 1'-0"

0 1" 2" 3" 6" 9" 1'

SCALE: 3" = 1'-0"

JOHNSON & ANDERSON, INC.
CONSULTING ENGINEERS
PONTIAC, MICHIGAN

REVISION	DATE	DESCRIPTION	BY

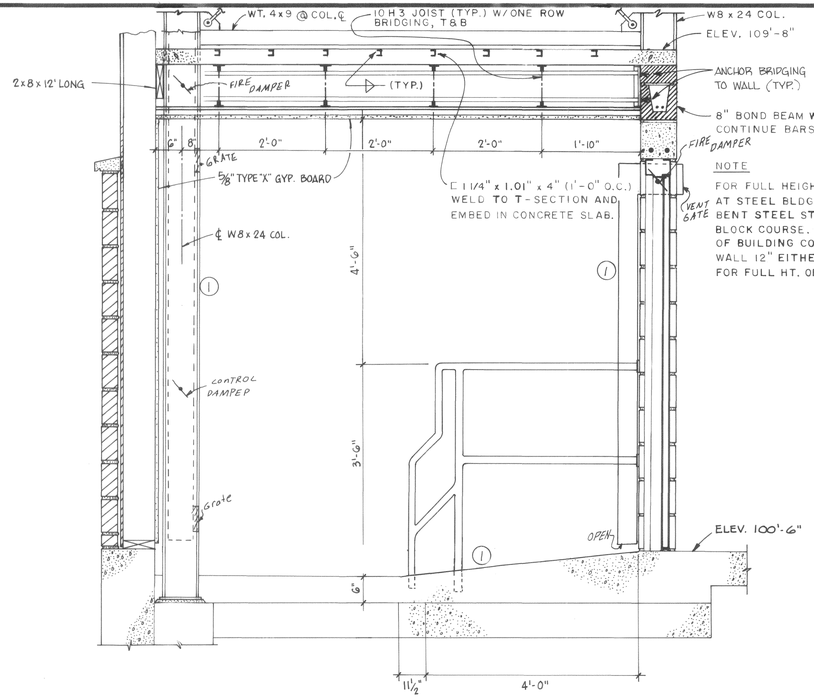
MAINTENANCE BUILDING SECTIONS & DETAILS

DESIGNED MAG | DRAWN W. DUNN DATE 5-22-80 | CHECKED DAC

DRAWING NO. 3R—MICH — 458 — 87

SHIAWASSEE NATIONAL WILDLIFE REFUGE

SHEET 14 OF 23



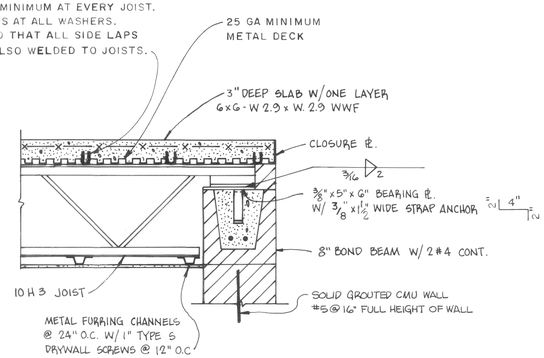
SECTION

SCALE: ¾" = 1'-0"

A 8 15

MEZZANINE OVER PAINT STORAGE. LIVE LOAD CAPACITY = 170 PSF

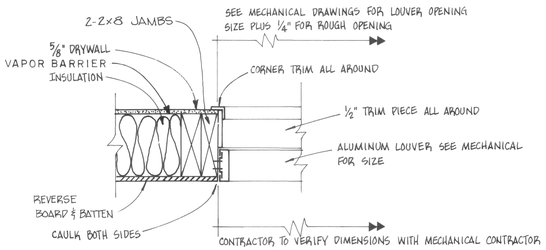
HIGH CHAIR WASHERS 10" O.C. MINIMUM AT EVERY JOIST. WELD THROUGH DECK TO JOISTS AT ALL WASHERS. WASHERS SHALL BE SPACED SO THAT ALL SIDE LAPS ARE WELDED TOGETHER AND ALSO WELDED TO JOISTS.



SECTION

SCALE: 1½" = 1'-0"

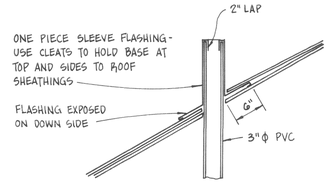
B 8 15



SECTION

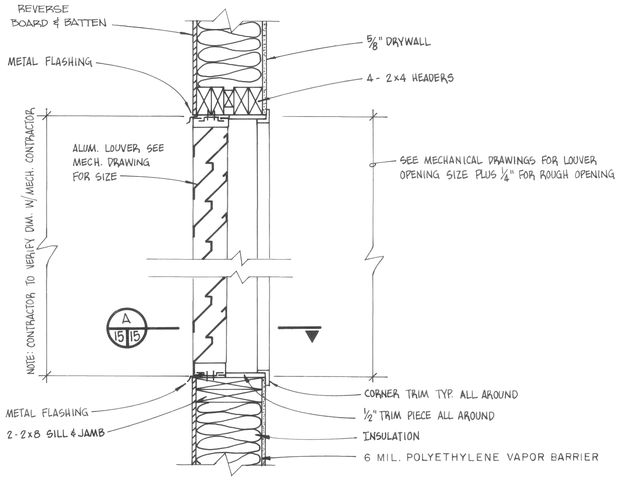
SCALE: 1½" = 1'-0"

A 15 15



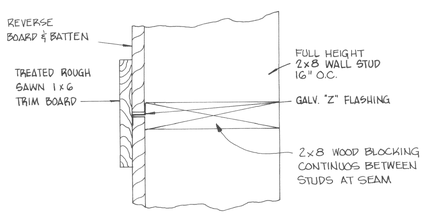
TYP. VENT PIPE THROUGH ROOF

NO SCALE



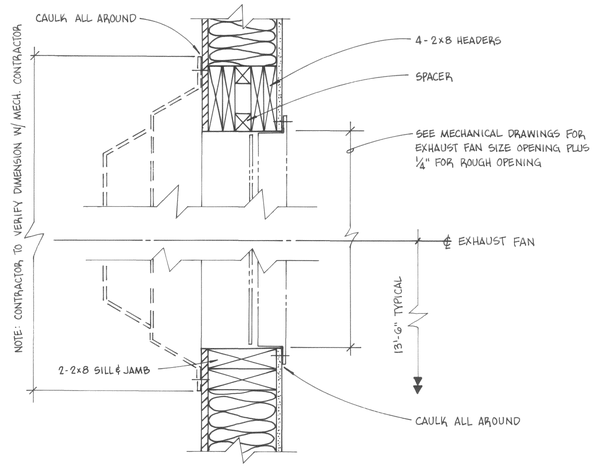
TYPICAL LOUVER DETAIL

SCALE: 1½" = 1'-0"



FLASH & TRIM BOARD DETAIL

NO SCALE



TYPICAL EXHAUST FAN OPENING DETAIL

SCALE: 1½" = 1'-0"

0 6" 1' 2' 4'

SCALE: ¾" = 1'-0"

0 3" 6" 9" 1' 2'

SCALE: 1½" = 1'-0"

JOHNSON & ANDERSON, INC.
CONSULTING ENGINEERS
PONTIAC, MICHIGAN

REVISION	DATE	DESCRIPTION	BY
1	9-9-83	"AS BUILT"	GAD

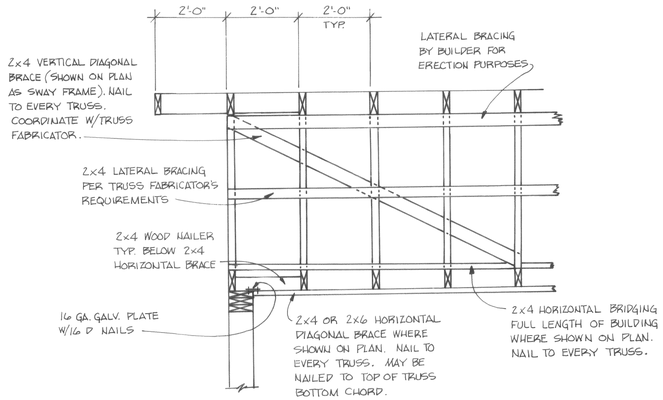
MAINTENANCE BUILDING SECTIONS & DETAILS

DESIGNED MAG. | DRAWN TDH W. DUNN | DATE 5-22-80 | CHECKED DKC

DRAWING NO. 3R — MICH. — 458 — 88

SHIAWASSEE NATIONAL WILDLIFE REFUGE

SHEET 15 OF 23



ROOF TRUSS BRACING (A 8/16)

ROOF BRACING NOTES (APPLIES TO ALL PERMANENT HORIZONTAL, DIAGONAL OR VERTICAL BRACING)

1.) USE 16D NAILS (TYP.) - 3½" LONG MIN.

2.) NUMBER OF NAILS (TYP.)
2x4 - TWO NAILS AT EACH TRUSS
2x6 - THREE NAILS AT EACH TRUSS

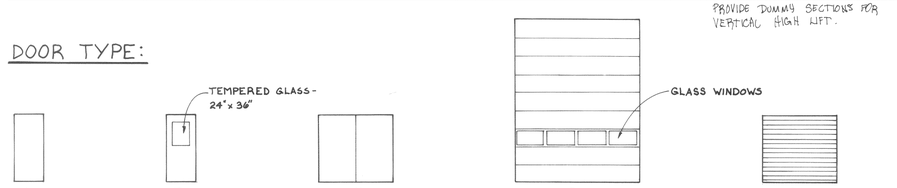
3) BRACING TO TOP PLATE CONNECTION (IF NAILED TO BOTTOM OF TRUSS BOTTOM CHORD)
2x4 - TWO NAILS INTO BRACE, TWO NAILS INTO TOP PLATE.
2x6 - THREE NAILS INTO BRACE, THREE NAILS INTO TOP PLATE.
(NOTE: AS ALTERNATE, TOOTHED GALVANIZED CONNECTOR PLATES OF EQUIVALENT STRENGTH TO ABOVE.) TO BE APPROVED BY OWNER'S REPRESENTATIVE.

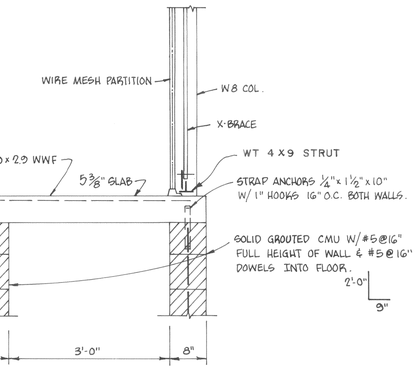
4) SIZE OF LATERAL BRACING FOR TRUSS WEB MEMBERS MAY VARY PER TRUSS FABRICATOR'S SPECIFICATIONS.

DOOR SCHEDULE

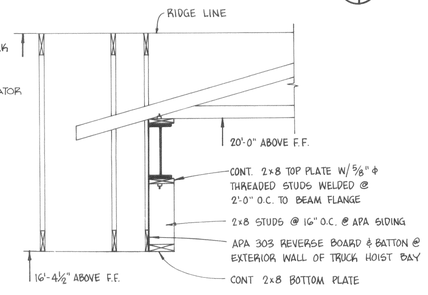
NO.	LOCATION	TYPE	SIZE	THICKNESS	MATERIAL DOOR	MATERIAL FRAME	DETAIL SHEET	SECT. LETTER	HDWE. SET	FIRE RATING	REMARKS
1	VESTIBULE	B	3'-0" x 7'-2"	1¾"	HOLLOW METAL	HOLLOW METAL	14	K,L,M		1 HR.	INSULATED
2	"	B	3'-0" x 7'-2"	"	" "	" "	14	K,L,M		"	"
3	OFFICE	A	3'-0" x 6'-8"	"	" "	" "	13	A,B,C		2 HR.	
4	TOILET	A	3'-0" x 6'-8"	"	" "	" "	13	A,B,C		"	UNDERCUT 1"
5	MECHANICAL ROOM	A	3'-0" x 6'-8"	"	" "	" "	13	A,B,C		"	
6	" "	A	4'-0" x 7'-2"	"	" "	" "	13	B,C,D		1 HR.	INSULATED
7	WELDING ROOM	C	8'-0 x 7'-2"	"	" "	" "	13	B,C,D		"	"
8	" "	A	3'-0 x 6'-8"	"	" "	" "	13	A,B,C		2 HR.	
9	CARPENTRY	C	8'-0 x 8'-0"	"	" "	" "	13	A,B,C		"	
10	BATTERY STORAGE	A	3'-0" x 6'-8"	"	" "	" "	13	A,B,C			"
11	JANITORS CLOSET	A	3'-0" x 6'-8"	"	" "	" "	13	A,B,C		"	
12	REPAIR AREA	B	3'-0" x 7'-2"	"	" "	" "	13	B,C,D		1 HR.	INSULATED
13	" "	D	14'-0" x 18'-0"	—	16 GA. STEEL	———	13	E,F,G		—	"
14	" "	D	14'-0" x 14'-0"	—	" " "	———	13	E,F,G		—	" SEE NOTE BELOW
15	PAINT, OIL, & GREASE STORAGE	A	3'-0" x 6'-8"	1¾"	HOLLOW METAL	HOLLOW METAL	13	A,B,C		2 HR.	
16	REPAIR AREA	D	14'-0" x 14'-0"	—	16 GA. STEEL	———	13	E,F,G		—	INSULATED
17	CARPENTRY	E	8'-0" x 6'-10"	—	24 GA. STEEL	STEEL	13	D,E,F		¾ HR	WEATHER STRIP
											NOTE: EXTEND TRACK TO CEILING, SAME AS DOOR 13. PROVIDE DUMMY SECTIONS FOR VERTICAL HIGH LIFT.

DOOR TYPE:

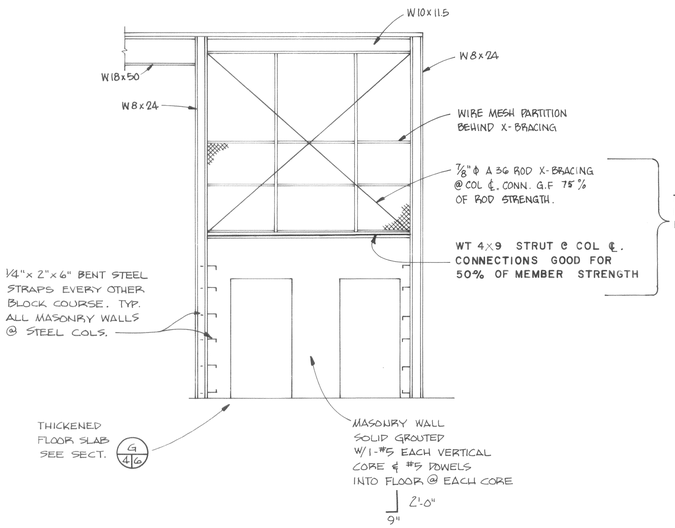




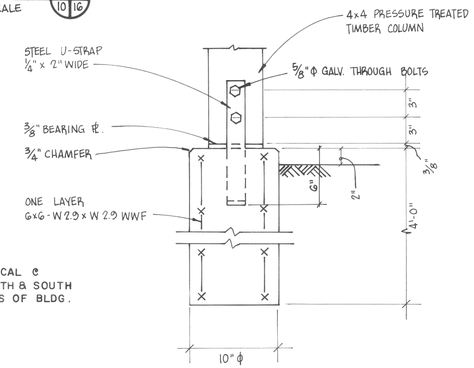
SECTION (E 10/16)
NO SCALE



SECTION (B 8/16)
NO SCALE



SECTION AT SHEAR WALL (F 8,10/16)
SCALE: ¼" = 1'-0"



SECTION (A 4/16)
SCALE: 1½" = 1'-0"

ROOM FINISH SCHEDULE

ROOM DESCRIPTION	FLOOR: CONCRETE	CERAMIC TILE	VINYL ASBESTOS	HARDNER	PLYWOOD	BASE: PAINTED MASONRY	RUBBER	CERAMIC TILE	WALL: MASONRY BLOCK	DRYWALL	8' HIGH HARDBOARD	CEIL.: 2 LAYER TYPE 'X' GYP. BD.	DRYWALL	REMARKS
VESTIBULE	•						•		•			•		
OFFICE			•				•		•			•	•	2 LAYERS TYPE "X" GYP. BD. FOR CEIL.
TOILET		•						•	•			•	•	" " " " "
MECHANICAL ROOM	•					•	•		•		•	•	•	" " " " "
WELDING ROOM	•					•	•		•			•	•	" " " " "
CARPENTRY ROOM	•					•	•		•			•	•	" " " " "
BATTERY STORAGE	•					•			•			•	•	" " " " "
JANITORS CLOSET	•					•			•			•	•	" " " " "
PAINT, OIL, & GREASE STORAGE	•					•	•		•			•	•	2 LAYERS TYPE "X" GYP. BD. FOR CEIL.
REPAIR AREA	•					•	•		•		•	•	•	
STORAGE MEZZANINE	•						•		•			•		

0 3" 6" 9" 1' 2'

SCALE: 1½" = 1'-0"

JOHNSON & ANDERSON, INC.
CONSULTING ENGINEERS
PONTIAC, MICHIGAN

REVISION	DATE	DESCRIPTION	BY

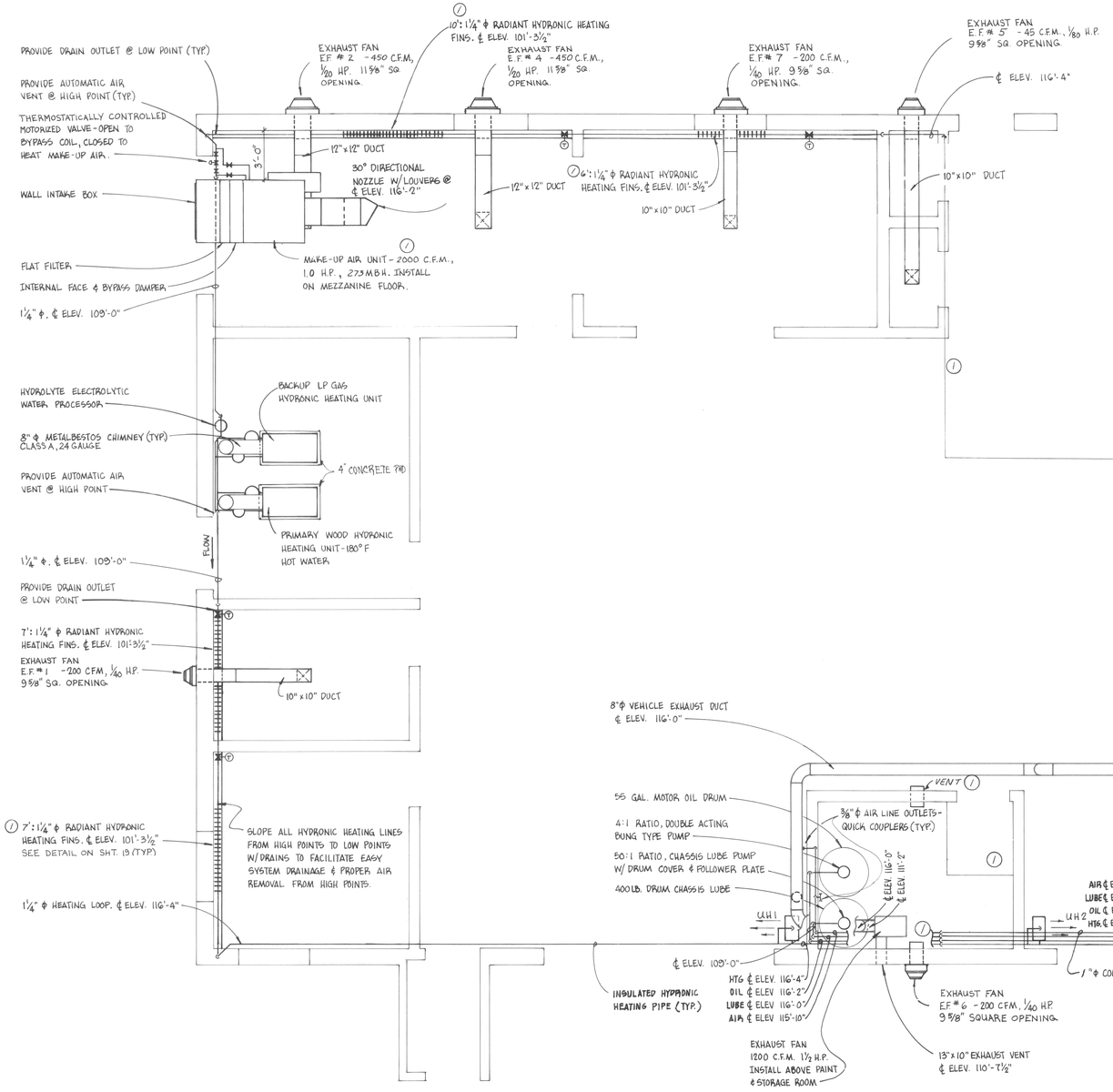
DOOR & FINISH SCHEDULE & MISCELLANEOUS DETAILS

DESIGNED JMC, MAG | DRAWN MAG | DATE 5-22-80 | CHECKED DAC

DRAWING NO. 3R— MICH. — 458 — 89

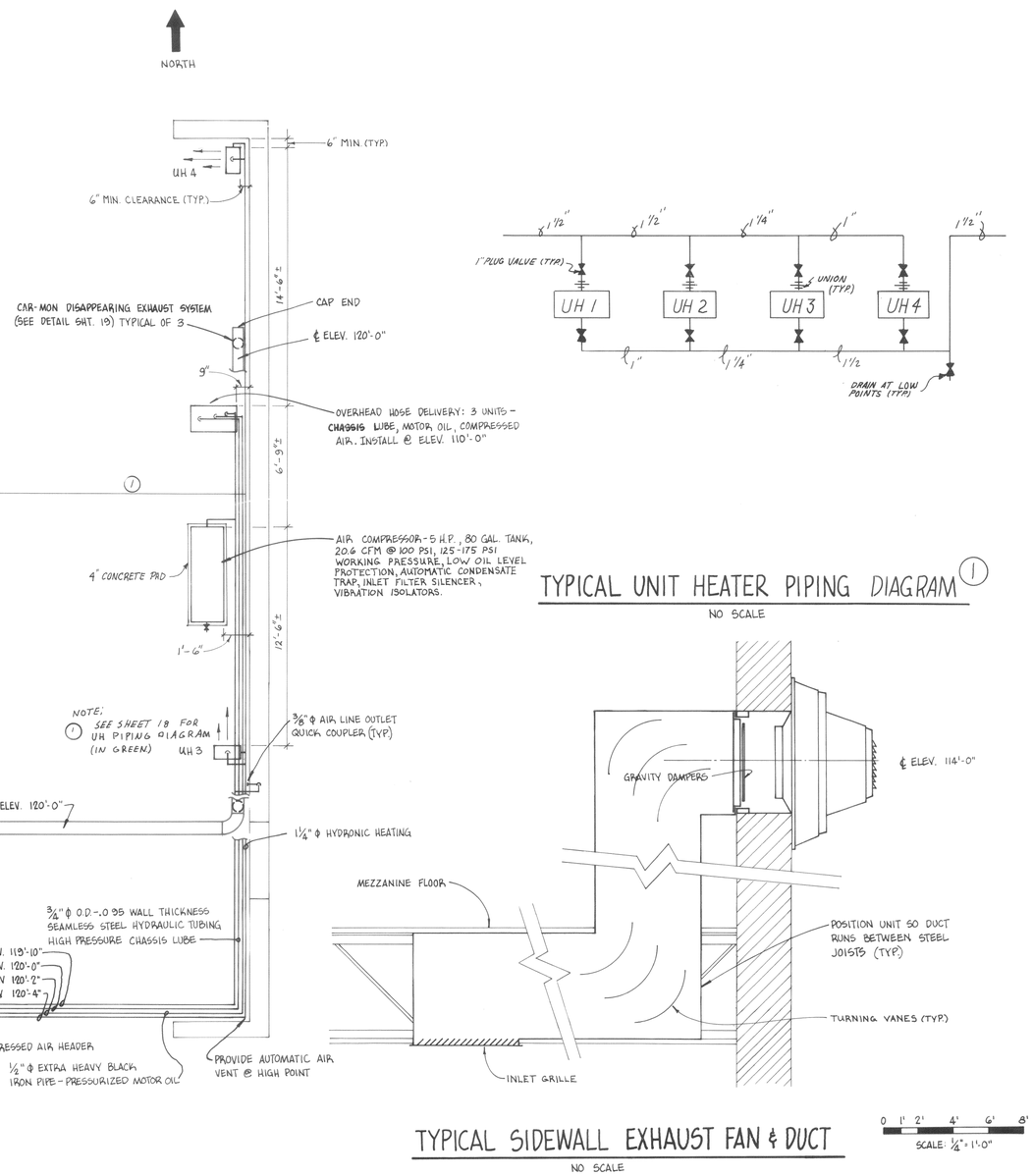
SHIAWASSEE NATIONAL WILDLIFE REFUGE

SHEET 16 OF 23



MAINTENANCE BUILDING - HEATING & VENTILATION LAYOUT

SCALE: ¼" = 1'-0"



TYPICAL UNIT HEATER PIPING DIAGRAM (1)

NO SCALE

TYPICAL SIDEWALL EXHAUST FAN & DUCT

NO SCALE

0 1' 2' 4' 6' 8'

SCALE: ¼" = 1'-0"

REVISION	DATE	DESCRIPTION	BY
(1)	8-31-83	"AS BUILT"	GAO

MAINTENANCE BUILDING - HEATING & VENTILATION LAYOUT

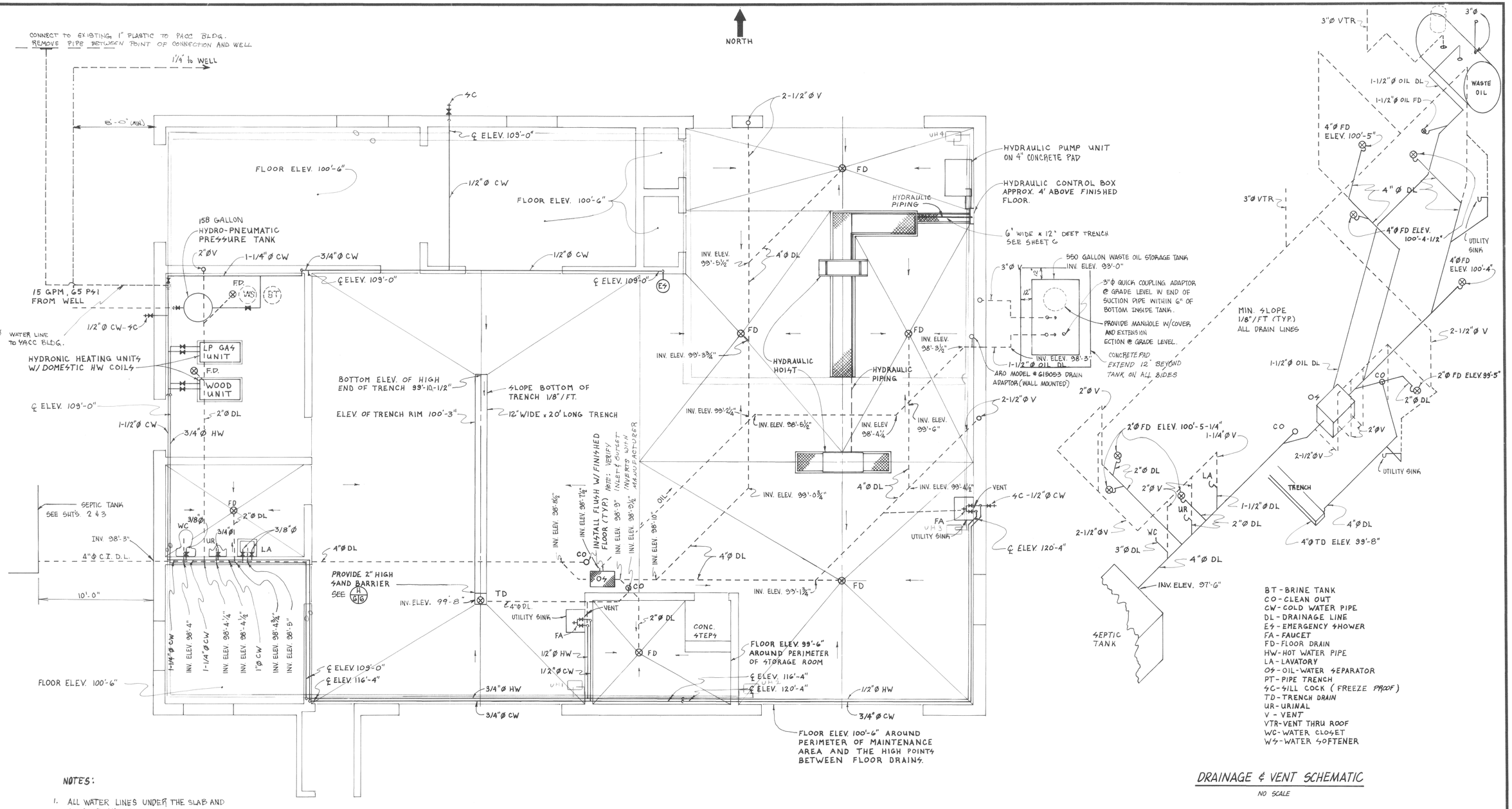
DESIGNED TDH | DRAWN W. DUNN DATE 5-22-80 | CHECKED DEF

DRAWING NO. 3R — MICH. — 458 — 90

JOHNSON & ANDERSON, INC.
CONSULTING ENGINEERS
PONTIAC, MICHIGAN

SHIAWASSEE NATIONAL WILDLIFE REFUGE

SHEET 17 OF 23



BT - BRINE TANK
CO - CLEAN OUT
CW - COLD WATER PIPE
DL - DRAINAGE LINE
ES - EMERGENCY SHOWER
FA - FAUCET
FD - FLOOR DRAIN
HW - HOT WATER PIPE
LA - LAVATORY
OS - OIL-WATER SEPARATOR
PT - PIPE TRENCH
SC - SILL COCK (FREEZE PROOF)
TD - TRENCH DRAIN
UR - URINAL
V - VENT
VTR - VENT THRU ROOF
WC - WATER CLOSET
WS - WATER SOFTENER

DRAINAGE & VENT SCHEMATIC

NO SCALE

NOTES:

1. ALL WATER LINES UNDER THE SLAB AND OUTSIDE THE BUILDING SHALL BE SDR 26 PVC PLASTIC.
2. DRAIN LINES UNDER THE SLAB AND OUT TO THE SEPTIC TANK AND WASTE OIL STORAGE TANK SHALL BE CAST IRON.

PLUMBING & DRAINAGE SYSTEM LAYOUT

SCALE: 1/4" = 1'-0"

0 1' 2' 4' 6' 8'
SCALE: 1/4" = 1'-0"

JOHNSON & ANDERSON, INC.
CONSULTING ENGINEERS
PONTIAC, MICHIGAN

REVISION	DATE	DESCRIPTION	BY
1	9-9-80	As Built	GJC

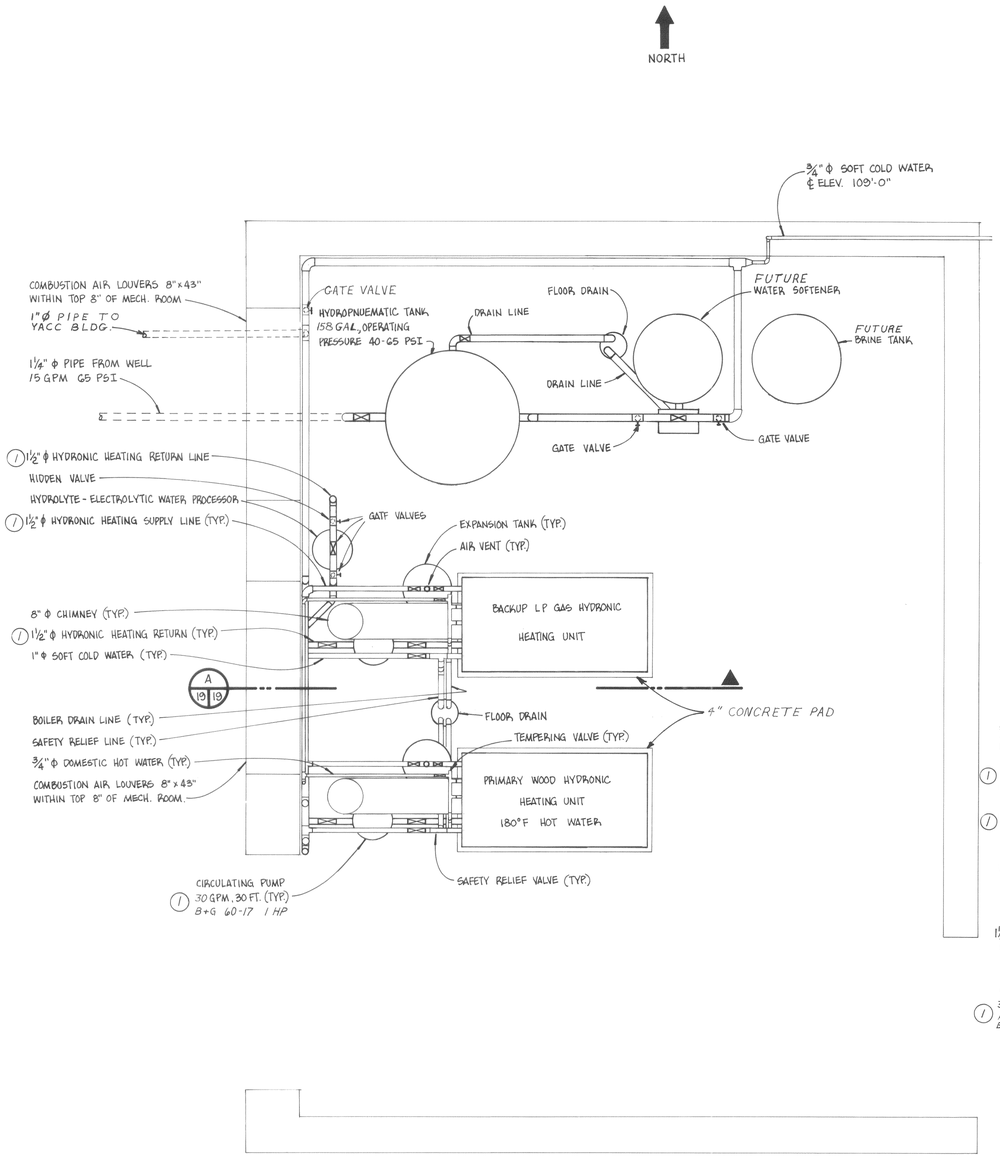
MAINTENANCE BUILDING PLUMBING & DRAINAGE

DESIGNED TDH | DRAWN TDH | DATE 5-22-80 | CHECKED DEF

DRAWING NO. 3R-MICH.-458-91

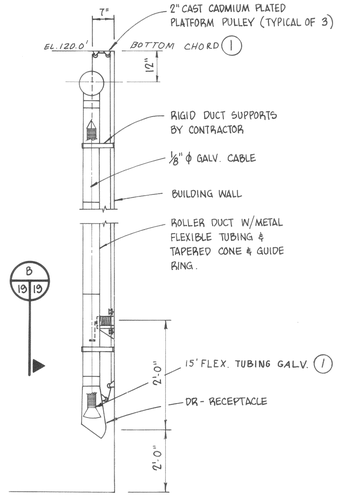
SHIAWASSEE NATIONAL WILDLIFE REFUGE

SHEET 18 OF 23



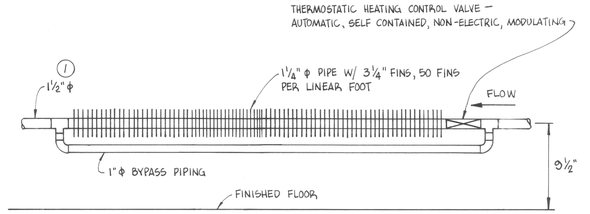
MECHANICAL ROOM - PLAN

SCALE: 3/4" = 1'-0"



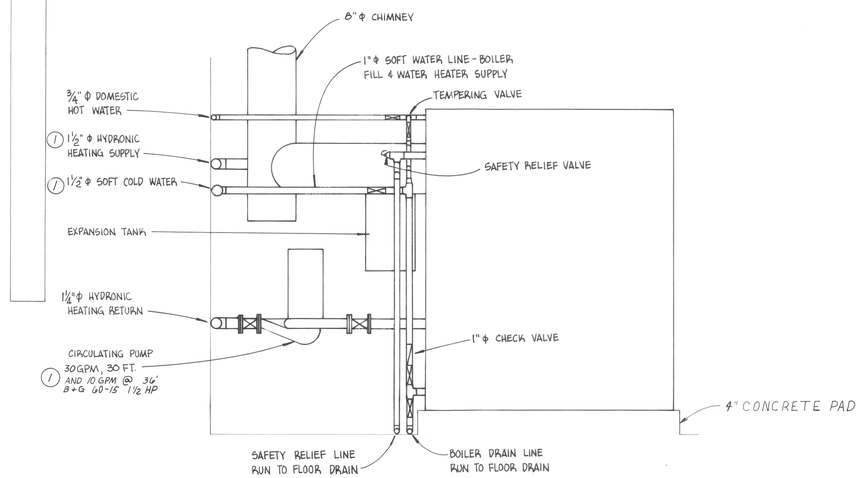
EXHAUST SYSTEM DETAIL

NO SCALE



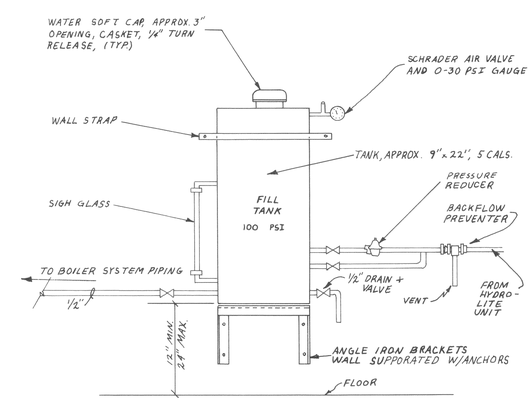
TYPICAL BASEBOARD HEATING

NO SCALE



BOILER PIPING A 19/19

SCALE: 1" = 1'-0"



(NO SCALE) (1)

0 6" 1' 2' 3' 5'
SCALE: 3/4" = 1'-0"

0 6" 1' 2' 3'
SCALE: 1" = 1'-0"

REVISION	DATE	DESCRIPTION	BY
(1)	9-/-83	"AS BUILT"	GAO

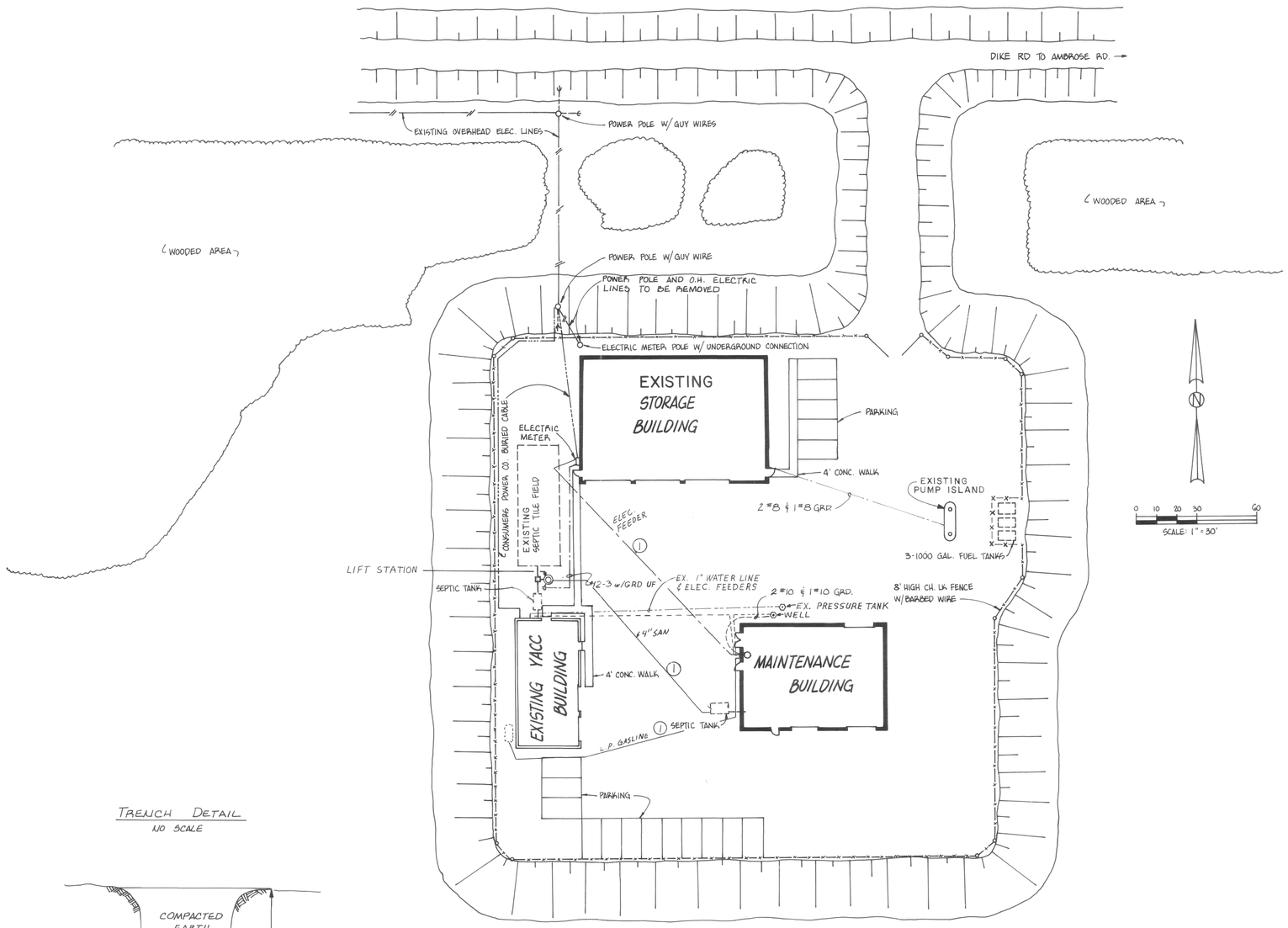
MECHANICAL ROOM - PLAN & DETAILS

DESIGNED TDH | DRAWN TDH | DATE 5-22-80 | CHECKED DEF

DRAWING NO. 3R— MICH — 458 — 92

JOHNSON & ANDERSON, INC.
CONSULTING ENGINEERS
PONTIAC, MICHIGAN

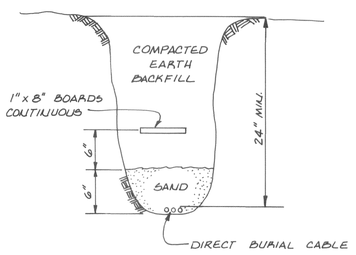
SHIAWASSEE NATIONAL WILDLIFE REFUGE



ELECTRICAL SYMBOL LIST

	FLUORESCENT LIGHTING FIXTURE
	LIGHTING FIXTURE, PENDANT OR CEILING MOUNTED
	LIGHTING FIXTURE, WALL MOUNTED
	120 V, 20 AMP, 1 Ø DUPLEX RECEPTACLE, GROUNDING TYPE
	WELDING RECEPTACLE
S	SINGLE POLE SWITCH
S_3	THREE-WAY SWITCH
	CIRCUIT BREAKER
	DISCONNECT SWITCH, FUSED
	CONTACTOR OR STARTER
T	THERMOSTAT
T	TRANSFORMER
P	PHOTOELECTRIC CELL
J	JUNCTION BOX
GRD	GROUND
	PANEL OR CABINET
WP	WEATHERPROOF, NEMA IV
EX PR	EXPLOSION PROOF
	HOME RUN - ARROWS INDICATE NO. OF CIRCUITS
	UNDER FLOOR CONDUIT
	CONDUIT BENT DOWN OR AWAY
	CONDUIT BENT UP OR TOWARD
—GC—	GROUND CONDUCTOR
	CONCEALED CONDUIT
25	3 PHASE MOTOR, HP AS INDICATED
1/6	1 PHASE MOTOR, HP AS INDICATED
	3 PHASE WELDING RECEPTACLE
SV	SOLENOID VALVE
UH	UNIT HEATER
	SEAL
P	PRESSURE SWITCH
	COMBINATION STARTER
	GROUND ROD
F	BUSS SBY-L 4" SQ, FUSE, SW., PILOT LIGHT

TRENCH DETAIL
NO SCALE



REVISION	DATE	DESCRIPTION	BY
1	9-1-83	"AS BUILT"	GAO

ELECTRICAL PLAN - SITE

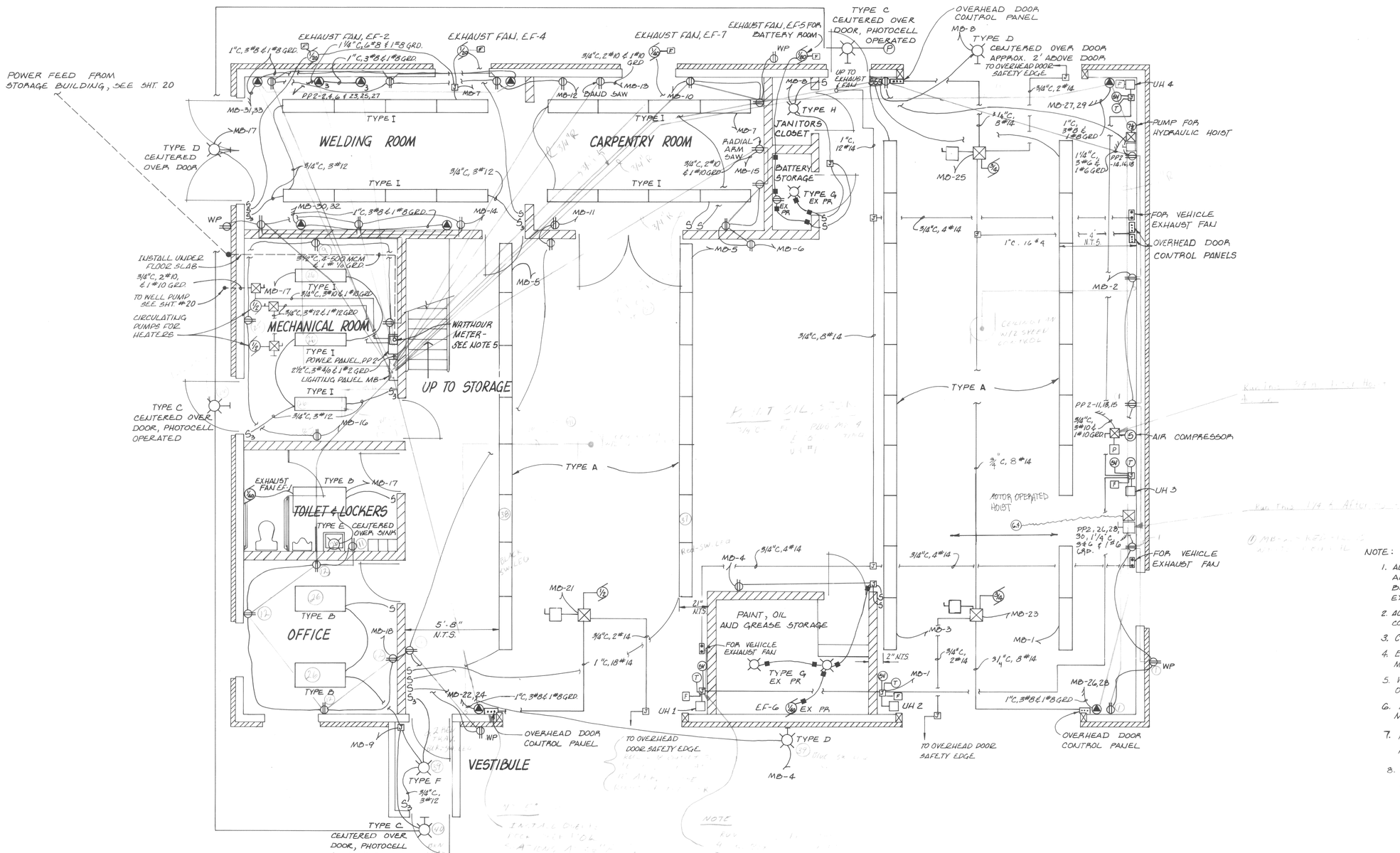
DESIGNED JAL | DRAWN W. DUNN DATE 5-22-80 | CHECKED DEF

DRAWING NO 3R—MICH—458—93

JOHNSON & ANDERSON, INC.
CONSULTING ENGINEERS
PONTIAC, MICHIGAN

SHIAWASSEE NATIONAL WILDLIFE REFUGE

SHEET 20 OF 23



ELECTRICAL PLAN - MAINTENANCE BUILDING

SCALE: 1/4" = 1'-0" 0' 1' 2' 4' 6' 8' 10'

NOTE:

1. ALL EQUIPMENT LOCATED IN THE PAINT, OIL, AND GREASE STORAGE ROOM OR THE BATTERY STORAGE ROOM SHALL BE EXPLOSION PROOF, CLASS 1, DIVISION 1.
2. ALL CIRCUITS SHALL BE 3/4" CONDUIT WITH 2 #12 CONDUCTORS UNLESS OTHERWISE NOTED.
3. CONDUIT TO BE ABOVE SUSPENDED CEILING.
4. ELECTRICAL SWITCHES AND OUTLETS TO BE A MINIMUM OF 18" ABOVE THE FLOOR.
5. WATTHOUR METER SHALL BE WESTINGHOUSE TYPE D4A-7 OR APPROVED EQUIVALENT. SUPPLIED BY CONTRACTOR.
6. ALL LIGHTING UNDER 8' CEILING TO BE SURFACE MOUNTED.
7. ALL DISCONNECT SWITCHES SHALL BE MOUNTED AT 6'-0" (TOP) ABOVE FIN. FLOOR.
8. ALL CONDUIT SHALL BE CONCEALED EXCEPT IN MECHANICAL, BATTERY STORAGE, AND PAINT OIL, AND GREASE STORAGE ROOMS.

REVISION	DATE	DESCRIPTION	BY
1	7-14-83	ADDED SHEET CONTROL	

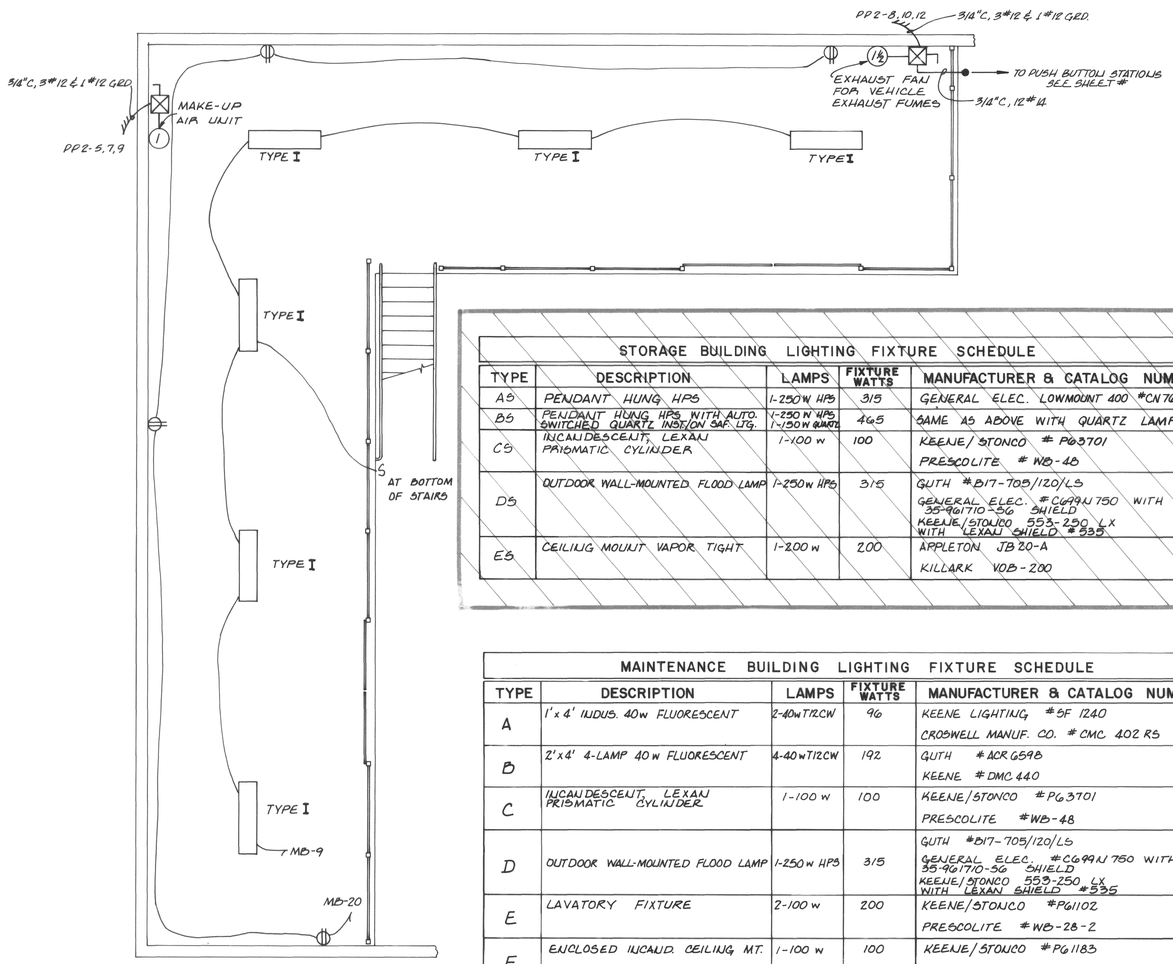
ELECTRICAL PLAN - MAINTENANCE BUILDING

DESIGNED JAL	DRAWN W. DUNN DATE 5-22-80	CHECKED
	DRAWING NO. 3R—MICH—458—94	

JOHNSON & ANDERSON, INC.
CONSULTING ENGINEERS
PONTIAC, MICHIGAN

SHIAWASSEE NATIONAL WILDLIFE REFUGE

SHEET 21 OF 23



STORAGE BUILDING

LIGHTING PANEL 'SBP' 120/240 V, 1 Ø, 3 W, WITH GROUND

100 A, 2P MAIN BREAKER

LOAD	TRIP	FRAME	POLE	EQUIPMENT	CKT	CKT	EQUIPMENT	POLE	FRAME	TRIP	LOAD
830	20	QOB	1	EXTERIOR LTG	1	2	RECEPTACLES	1	QOB	20	1000
1410	20	QOB	1	LIGHTING	3	4	RECEPTACLES	1	QOB	20	1000
1810	20	QOB	1	LIGHTING	5	6	RECEPTACLES	1	QOB	20	800
	40	QOB	2	WELDING RECEPTACLE - NORTH WALL	7 9	8 10	WELDING RECEPTACLE - WEST WALL	2	QOB	40	
828	20	QOB	1	OVERHEAD DOOR	11	12	OVERHEAD DOOR	1	QOB	20	828
828	20	QOB	1	OVERHEAD DOOR	13	14	FUEL PUMPS	1	QOB	30	2052
					15	16					

NOT IN THIS CONTRACT

STORAGE BUILDING LIGHTING FIXTURE SCHEDULE

TYPE	DESCRIPTION	LAMPS	FIXTURE WATTS	MANUFACTURER & CATALOG NUMBER
A5	PENDANT HUNG HPS	1-250W HPS	315	GENERAL ELEC. LOWMOUNT 400 #CN76BN303
B5	PENDANT HUNG HPS WITH AUTO. SWITCHED QUARTZ INSTAION SAF. LTG.	1-250 W HPS 1-150 W QUARTZ	465	SAME AS ABOVE WITH QUARTZ LAMP
C5	INCANDESCENT, LEXAN PRISMATIC CYLINDER	1-100 w	100	KEENE/STONCO # PG3701 PRESCOLITE # WB-48
D5	OUTDOOR WALL-MOUNTED FLOOD LAMP	1-250w HPS	315	GUTH #B17-705/120/LS GENERAL ELEC. #CG99NJ750 WITH 35-961710-56 SHIELD KEENE/STONCO 553-250 LX WITH LEXAN SHIELD #535
E5	CEILING MOUNT VAPOR TIGHT	1-200 w	200	APPLETON JB 20-A KILLARK VOB-200

MAINTENANCE BUILDING LIGHTING FIXTURE SCHEDULE

TYPE	DESCRIPTION	LAMPS	FIXTURE WATTS	MANUFACTURER & CATALOG NUMBER
A	1'x4' INDUS. 40w FLUORESCENT	2-40WT12CW	96	KEENE LIGHTING #SF 1240 CROSWELL MANUF. CO. # CMC 402 RS
B	2'x4' 4-LAMP 40w FLUORESCENT	4-40WT12CW	192	GUTH #ACR G598 KEENE #DMC 440
C	INCANDESCENT, LEXAN PRISMATIC CYLINDER	1-100 w	100	KEENE/STONCO #PG3701 PRESCOLITE #WB-48
D	OUTDOOR WALL-MOUNTED FLOOD LAMP	1-250w HPS	315	GUTH #B17-705/120/LS GENERAL ELEC. #CG99NJ750 WITH 35-961710-56 SHIELD KEENE/STONCO 553-250 LX WITH LEXAN SHIELD #535
E	LAVATORY FIXTURE	2-100 w	200	KEENE/STONCO #PG1102 PRESCOLITE #WB-28-2
F	ENCLOSED INCAND. CEILING MT.	1-100 w	100	KEENE/STONCO #PG1183 PRESCOLITE #CF-48
G	EXPLOSION-PROOF, CLASS 1, DIVISION 1, GROUP A, INCAN/DESCENT	300w PS30	300	CROUSE-HINDS #EVC.X282 KILLARK #EXG-2-200
H	CEILING MOUNT, VAPOR TIGHT	1-200 w	200	APPLETON JB 20-A KILLARK VOB-200
I	1'x4' SURFACE MOUNT, ENCLOSED	2-40WT12CW	96	KEENE #SCC240 GUTH #VP6762

STORAGE BUILDING

POWER PANEL 'PP1' 120/240 V, 3 Ø, 3 W, WITH GROUND

400 A, 3P MAIN BREAKER

LOAD	TRIP	FRAME	POLE	EQUIPMENT	CKT	CKT	EQUIPMENT	POLE	FRAME	TRIP	LOAD
					1	2					
3.6A	20	FA	3	EXHAUST FAN EF-1	3	4	SPARE	3	FA	20	
					5	6					
					7	8					
332A	350	LA	3	MAINTENANCE BUILDING	9	10	PANEL SDP	2	FA	100	65A
					11	12	ALARM LIGHT	1	QOB	10	
					13	14	SEWAGE PUMP	1	QOB	15	8A
3.6A	20	FA	3	EXHAUST FAN EF-2	15	16	SPARE				
					17	18					

NEUTRAL

ELECTRICAL PLAN - STORAGE MEZZANINE
MAINTENANCE BUILDING

SCALE: 1/4" = 1'-0" 0' 1' 2' 4' 6' 8' 10'

JOHNSON & ANDERSON, INC.
CONSULTING ENGINEERS
PONTIAC, MICHIGAN

REVISION	DATE	DESCRIPTION	BY

ELECTRICAL PLAN - STORAGE MEZZANINE

DESIGNED JAL | DRAWN W. DUNN DATE 5-22-80 | CHECKED DEF

DRAWING NO. 3R — MICH. — 458 — 95

SHIAWASSEE NATIONAL WILDLIFE REFUGE

MAINTENANCE BUILDING

LIGHTING PANEL 'MB' 120/240 V, 1 Ø, 3 W, WITH GROUND

225 A, 2P MAIN BREAKER

LOAD	TRIP	FRAME	POLE	EQUIPMENT	CKT	CKT	EQUIPMENT	POLE	FRAME	TRIP	LOAD
1280	20	QOB	1	HIGH BAY LIGHTING & UNIT HEATERS	1	2	RECEPTACLES - EAST WALL	1	QOB	20	1000
960	20	QOB	1	HIGH BAY LIGHTING	3	4	PAINT STORAGE ROOM	1	QOB	20	1179
1728	20	QOB	1	LOW BAY LIGHTING	5	6	RECEPTACLES - CARPENTRY	1	QOB	20	600
1656	20	QOB	1	CARPENTRY & WELDING LTG	7	8	BATTERY STORAGE ROOM & JANITOR'S CLOSET & FAN	1	QOB	20	1071
1360	20	QOB	1	OFFICE, STORAGE, MEZZANINE & OTHER LIGHTING	9	10	RECEPTACLES - CARPENTRY	1	QOB	20	660
800	20	QOB	1	RECEPTACLES - LOW BAY	11	12	RECEPTACLES - WELDING	1	QOB	20	800
2040	30	QOB	1	BAND SAW	13	14	RECEPTACLES - WELDING	1	QOB	20	800
2040	30	QOB	1	RADIAL ARM SAW	15	16	RECEPTACLES - MECHANICAL RM.	1	QOB	20	800
1059	20	QOB	1	MECHANICAL & LAV. LTG	17	18	RECEPTACLES - OFFICE	1	QOB	20	1000
				SPACE	19	20	RECEPTACLES - STORAGE MEZZ.	1	QOB	20	800
588	20	QOB	1	OVERHEAD DOOR	21	22	WELDER - LOW BAY	2	QOB	40	32A
828	20	QOB	1	OVERHEAD DOOR	23	24					
828	20	QOB	1	OVERHEAD DOOR	25	26	WELDER - HIGH BAY	2	QOB	40	32A
32A	40	QOB	2	WELDER - HIGH BAY	27	28					
					29	30	WELDING - WELDING RM	2	QOB	40	32A
32A	40	QOB	2	WELDER - WELDING ROOM	31	32					
					33	34	CIRCULATING PUMP	1	QOB	20	1120
				SPACE	35	36					
				SPACE	37	38	SPACE				
				SPACE	39	40	SPACE				
				SPACE	41	42	SPACE				

NEUTRAL

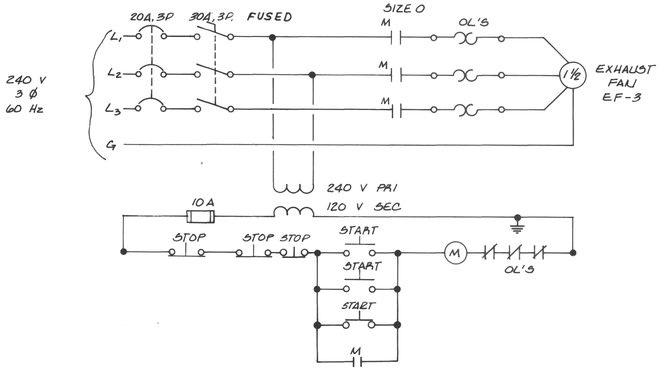
MAINTENANCE BUILDING

POWER PANEL 'PP2' 120/240 V, 3 Ø, 3 W, WITH GROUND (42 SPACES)

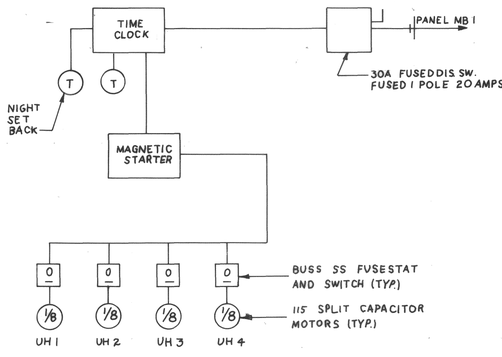
350 A, 3P MAIN BREAKER

LOAD	TRIP	FRAME	POLE	EQUIPMENT	CKT	CKT	EQUIPMENT	POLE	FRAME	TRIP	LOAD
223A	225	KA	2	PANEL MB	1	2					
					3	4	HOLBART WELDER	3	FA	40	
					5	6					
3.6A	20	FA	3	MAKE-UP AIR FAN EF-B	7	8					
					9	10	VEHICLE EXHAUST FAN EF-3	3	FA	20	52A
					11	12					
15.2A	30	FA	3	AIR COMPRESSOR	13	14					
					15	16	HYDRAULIC HOIST	3	FA	50	22A
					17	18					
				SPACE	19	20					
					21	22	SPACE				
					23	24					
5.8A	20	FA	3	HACKSAW	25	26					
					27	28	MOTOR OPERATED HOIST	3	FA	30	19
8.0A	20	FA	2	WATER PUMP	29	30					
					31	32	SPACE				
				SPACE	39	40	SPACE				
				SPACE	41	42	SPACE				

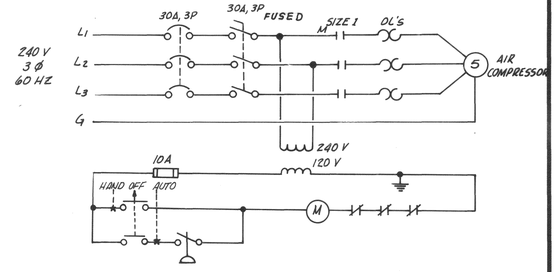
NEUTRAL



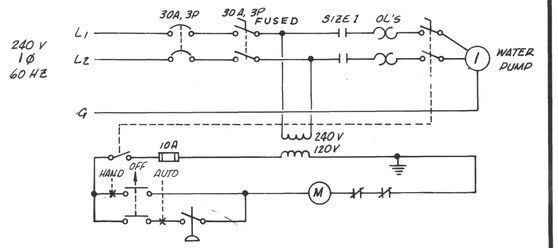
SCHEMATIC DIAGRAM
VEH. EXHAUST FAN EF-3



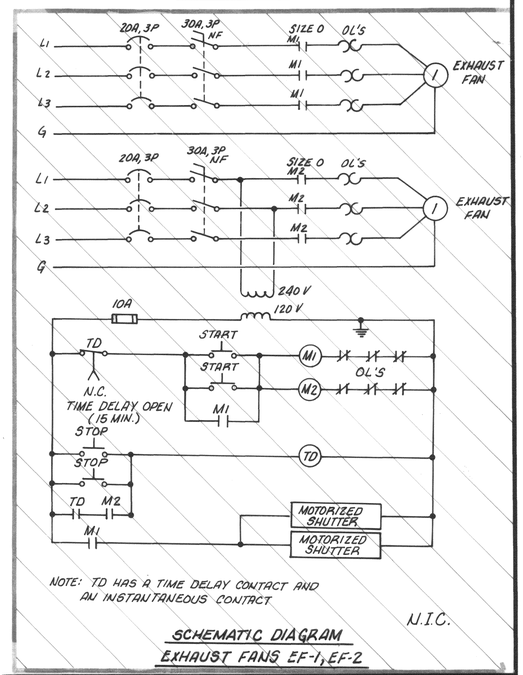
ELECTRIC DIAGRAM



SCHEMATIC DIAGRAM
AIR COMPRESSOR



SCHEMATIC DIAGRAM
WATER PUMP



NOTE: TD HAS A TIME DELAY CONTACT AND AN INSTANTANEOUS CONTACT

N.I.C.

SCHEMATIC DIAGRAM
EXHAUST FANS EF-1, EF-2

REVISION	DATE	DESCRIPTION	BY

ELECTRICAL PLAN - SCHEMATICS

DESIGNED JAL | DRAWN JHM | DATE 5-22-80 | CHECKED DEF

DRAWING NO. 3R — MICH — 458 — 96

JOHNSON & ANDERSON, INC.
CONSULTING ENGINEERS
PONTIAC, MICHIGAN

SHIAWASSEE NATIONAL WILDLIFE REFUGE

SHEET 23 OF 23